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# Mdisp functions

## Synopsis

The functions prefixed with Mdisp make up the Display module. The Display module allows you to display images and manipulate their display. The Display module offers many display effects, such as annotation, LUTs, and panning and zooming.

## Functions

MdispAlloc
MdispControl
MdispFree
MdispGetHookInfo
MdispHookFunction
MdispInquire
MdispLut
MdispPan
MdispSelect
MdispSelectWindow
MdispZoom

# MdispAlloc

## Synopsis

Allocate a display.

## Syntax

```
AIL_ID MdispAlloc(
    AIL_ID SystemId,               //in
    AIL_INT DispNum,               //in
    AIL_CONST_TEXT_PTR DispFormat, //in
    AIL_INT64 InitFlag,            //in
    AIL_ID *DisplayIdPtr           //out
)
```

## Description

This function allocates a display on the specified system so that it can be used by subsequent Aurora Imaging Library display functions. Use MdispSelect() to select the image buffer to display. Note that the buffer and the display should be allocated on the same system.

In general, the display is presented on the computer running the main Aurora Imaging Library application. However, if you are developing a Distributed Aurora Imaging Library application or an Aurora Imaging Web Library server application, you can allocate a display to be shown on a remote computer.

If you are developing a Distributed Aurora Imaging Library controlling application and you allocate a display on a Distributed Aurora Imaging Library remote system, you can select image buffers allocated on that remote system to the display. By default, these buffers are displayed on the local computer. However, to display the buffers on the remote computer, combine M_EXCLUSIVE or M_WINDOWED with M_REMOTE_DISPLAY when allocating the display. This type of display is referred to as a remote display.

If you are developing an Aurora Imaging Library application that also acts as an Aurora Imaging Web Library server, you can allocate a display that can be presented in one or more instances of an Aurora Imaging Web Library client application (for example, running in a compatible web browser). To do so, use M_WEB when allocating the display. This type of display is referred to as an Aurora Imaging Web Library display.

After allocating the display, you should check if the operation was successful, using MappGetError() or by verifying that the display identifier returned is not **M_NULL** (or *nullptr* if M_UNIQUE_ID was specified).

When the display is no longer required, release it using MdispFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the display's lifetime and you must not manually free it.

## Parameters

SystemId INQ

Specifies the system on which to allocate the display. This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For specifying the system | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| M_DEFAULT_HOST | Specifies the default Host system of the current Aurora Imaging Library application. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *System identifier* | Specifies a valid system identifier, previously allocated using MsysAlloc(). | c | | e | f | g | a | | i | | j | k | l | m | n | b |

DispNum INQ

Specifies the device to use for image display.

For a non-exclusive display, or for an exclusive display that you want Aurora Imaging Library to choose the display device, set this parameter to the following value:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For specifying the display device | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| M_DEFAULT | Specifies that Aurora Imaging Library will use the most appropriate device for display purposes. If your imaging board has a display section, and it is available, Aurora Imaging Library will typically use it for display purposes. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

For an exclusive display for which you want to specify the position of the display device in screen arrangements of up to 3x3 screens, this parameter can be set to a combination of the following values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For specifying the position of the device to use for an exclusive display | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| M_BOTTOM | Specifies to use the bottom-most device for an exclusive display. Note that this option cannot be combined with M_TOP. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_CENTER | Specifies to use the center device for an exclusive display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_LEFT | Specifies to use the left-most device for an exclusive display. Note that this option cannot be combined with M_RIGHT. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_RIGHT | Specifies to use the right-most device for an exclusive display. Note that this option cannot be combined with M_LEFT. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_TOP | Specifies to use the top-most device for an exclusive display. Note that this option cannot be combined with M_BOTTOM. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

DispFormat INQ

Specifies the format of the display. For windowed displays, this parameter has no effect and should be set to "M_DEFAULT". For exclusive displays, this parameter specifies the screen resolution.

For any type of display, this parameter can be set to the following value:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For specifying the default display format | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| AIL_TEXT("M_DEFAULT") [1] | Specifies the default display format. For windowed displays, this is the only supported setting. For exclusive displays, this specifies the default video configuration format (VCF), which can be set during installation and later modified using the Aurora Imaging Configurator utility (Defaults tab). | c | | e | f | g | a | | i | | j | k | l | m | n | b |

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

For exclusive displays, the DispFormat parameter can be set to the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For exclusive displays | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| M_CURRENT_RESOLUTION | Specifies to use the current resolution of the screen. This option is only available for an M_EXCLUSIVE display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| AIL_TEXT("*vcf-file-name.vcf*") [1] | Specifies the name of the video configuration format (VCF) that defines the resolution and refresh rate to use. The file typically has a VCF extension. The drive, directory, and name of the file must be specified, for example: "*C:\mydirectory\myfile*".<br><br>For a file on the remote computer (under Distributed Aurora Imaging Library), the path and file name must be preceded by "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").<br><br>The file name string can be specified in, for example, the "1024x768x32@60hz.vcf" format. For the current list of video configuration format (VCF) files, see the installed VCF folder, for example, *C:\Program Files\Aurora Imaging Library\11\Drivers\Host\vcf*. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

InitFlag INQ

Specifies your display's type. This parameter can be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For specifying the display type | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| M_DEFAULT | Specifies the default display type, which can be set using the Aurora Imaging Configurator utility. Typically, the default display type is M_WINDOWED. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_EXCLUSIVE + | Specifies to present the image buffer selected for display full-screen, without a windowed border, in one of Windows desktop screens. The image buffer is presented at the center of the screen.<br><br>To use an exclusive display, your Windows desktop should be using more than one screen. Allocating an exclusive display on the main screen might not be convenient; if you do so, the Aurora Imaging Library exclusive display will hide third-party applications designed to start on the main screen and hide the Windows taskbar.<br><br>> This display type requires MdispSelect() to display images. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_WEB | Specifies to present the image buffer selected for display in one or more instances of an Aurora Imaging Web Library client application (for example, running in a compatible web browser).<br><br>To use an Aurora Imaging Web Library display, you must enable Aurora Imaging Web Library server functionality in your application using MappControl() with M_WEB_CONNECTION. You must also publish the display using MobjControl() with M_WEB_PUBLISH.<br><br>> An Aurora Imaging Web Library display cannot be presented by your Aurora Imaging Library application. It can only be presented by a connected Aurora Imaging Web Library client application (running on the local computer or a remote computer).<br><br>> This display type requires MdispSelect() to display images. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOWED + | Specifies to present the image buffer selected for display in its own window on the Windows desktop screen(s). The display window is tracked and updated with the image buffer selected for display; that is, if the window moves or is occluded, the window is updated with the image buffer accordingly.<br><br>> This display type can be used with MdispSelect() and MdispSelectWindow() to display images, unless combined with M_REMOTE_DISPLAY. A remote display requires MdispSelect() to display images. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WPF | Specifies to present the image buffer selected for display using an AILWPFDisplay control. See the Using WPF with Aurora Imaging Library section of Chapter 60: Using Aurora Imaging Library with .NET in the User Guide for details of how to use.<br><br>> This display type requires MdispSelect() to display images. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

Combination value for M_EXCLUSIVE; M_WINDOWED.

You can add the following value to the above-mentioned values to set where and how the display is allocated.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For controlling where and how the display is allocated | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Combination value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_REMOTE_DISPLAY | Specifies that the display is displayed on the remote computer in a Distributed Aurora Imaging Library application. This setting is only available when the display is allocated on a remote system. If the display is allocated on a remote system but this setting is not specified, the display is presented on the local computer.<br><br>To present the display on the remote computer, the Distributed Aurora Imaging Library server cannot be running as a service; you must start it manually. To do so, assuming that Distributed Aurora Imaging Library has already been installed during the setup, you must logon to the remote computer and, from the Aurora Imaging Configurator utility, push the **Start Server** button, found in the **Server Settings** pane under the **Distributed AIL** item. For more information, see the Setting up the Distributed Aurora Imaging Library server on remote computers subsection of the Preparing computers for Distributed Aurora Imaging Library section of Chapter 63: Distributed Aurora Imaging Library.<br><br>> This display type requires MdispSelect() to display images. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

DisplayIdPtr

Specifies the address of the variable in which to write the display identifier or specifies the data type that the function should use to return the display identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For retrieving the identifier or specifying how to return it | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| M_NULL | Specifies that you will use this function's return value to obtain the identifier of the allocated display ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_UNIQUE_ID | Specifies that you will use this function's return value to obtain the identifier of the allocated display ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_DISP_ID* is returned instead of a standard Aurora Imaging Library identifier.<br><br>> This setting is only available when using C++11 (or later).<br><br>An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the display (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Address for the display identifier* | Specifies the address of an *AIL_ID* in which to write the identifier of the allocated display.<br><br>If allocation fails, **M_NULL** is written as the identifier. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

## Return value

The returned value is the display identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_DISP_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

## Compilation information

| | |
|---|---|
| **Header** | Include AIL.h. |
| **Library** | Use ail11.lib. |
| **DLL** | Requires ail11.dll; aildisplay11.dll. |

# MdispControl

## Synopsis

Control a display setting.

## Syntax

```
void MdispControl(
    AIL_ID DisplayId,       //out
    AIL_INT64 ControlType,  //in
    AIL_DOUBLE ControlValue //in
)
```

## Description

This function allows you to control the specified Aurora Imaging Library display setting.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MdispControl()** internally calls the *AIL_INT64* version of this function (MdispControlInt64()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MdispControlDouble()).

## Parameters

DisplayId

Specifies the identifier of the target display.

ControlType

Specifies the type of display setting to control. The control types for windowed displays can control the default Aurora Imaging Library or user-specified window of the display (MdispSelect() or MdispSelectWindow()).

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the new value to assign to the display setting specified by the ControlType parameter.

See the Parameter associations section for possible values that can be specified.

## Parameter associations

The tables below list possible values for the **ControlType** and **ControlValue** parameters.

- For all types of displays
- For windowed displays
- For Aurora Imaging Library windowed displays
- For user windowed displays
- For displays allocated on a Distributed Aurora Imaging Library remote system but displayed on the local computer

Unless otherwise specified, the following ControlType and corresponding ControlValue parameter settings are available for all types of displays.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For all types of displays

| ControlType / ControlValue | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_ASSOCIATED_GRAPHIC_LIST_ID | Sets the 2D graphics list to associate with the display. Only one 2D graphics list can be associated with the display at a time. If you associate a 2D graphics list with a display that already has been associated with one, the previous 2D graphics list is disassociated from the display and the new one is associated. INQ | | | | | | | | | | | | | | | |
| M_NULL | Specifies that no 2D graphics list is associated with the display.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *2D graphics list identifier* | Specifies the identifier of the 2D graphics list to associate with the display. The 2D graphics list must have been previously allocated using MgraAllocList(). In this case, the vector-based graphics, contained within the 2D graphics list, will be used to annotate the display non-destructively.<br><br>By default, modifications to the graphics within the list (added, deleted, or altered graphics) will be immediately reflected in the non-destructive annotation of the display. To change this behavior, use M_UPDATE_GRAPHIC_LIST.<br><br>To release the 2D graphics list from memory, use MgraFree(). If the associated 2D graphics list is freed, it is automatically disassociated from the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_BACKGROUND_COLOR | Sets the background color for the display. If the selected image buffer is smaller than the display or does not fill the display completely, the surrounding area will be displayed with this color. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DEFAULT | Same as M_COLOR_WHITE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_RGB888(<br>AIL_INT Red,<br>AIL_INT Green,<br>AIL_INT Blue<br>) | Specifies the RGB value to which the background color will be set. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | **Parameters** | | | | | | | | | | | | | | | |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Red<br><br>Specifies the red band, as a value between 0 and 255. | | | | | | | | | | | | | | |
| | | Green<br><br>Specifies the green band, as a value between 0 and 255. | | | | | | | | | | | | | | |
| | | Blue<br><br>Specifies the blue band, as a value between 0 and 255. | | | | | | | | | | | | | | |
| | M_COLOR_BLACK | Specifies the color black. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BLUE | Specifies the color blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BRIGHT_GRAY | Specifies the color bright gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_CYAN | Specifies the color cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_BLUE | Specifies the color dark blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_CYAN | Specifies the color dark cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_GREEN | Specifies the color dark green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_MAGENTA | Specifies the color dark magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_RED | Specifies the color dark red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_YELLOW | Specifies the color dark yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GRAY | Specifies the color gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GREEN | Specifies the color green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_BLUE | Specifies the color light blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GRAY | Specifies the color light gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GREEN | Specifies the color light green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_COLOR_LIGHT_WHITE | Specifies the color light white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_MAGENTA | Specifies the color magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_RED | Specifies the color red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_WHITE | Specifies the color white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_YELLOW | Specifies the color yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_CENTER_DISPLAY | | Sets whether a selected image buffer will be centered in the display, both in X and Y. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Specifies the default value. The default value is M_ENABLE for an Aurora Imaging Library default window and an exclusive display. The default value is M_DISABLE for a user-defined window display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that the image buffer will not be centered in the display. This will overwrite the current panning offsets (MdispPan()) to (0,0). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that the image buffer will always be centered in the display, both in X and Y. This will offset the image buffer by ( (WindowSizeX - BufferSizeX)/2, (WindowSizeY - BufferSizeY)/2) in the window, and current panning offsets (MdispPan()) are overwritten with these values. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_GRAPHIC_LIST_INTERACTIVE | | Sets whether a user can interactively modify the graphics in the 2D graphics list associated with the display. Modifications to the graphics in the list include moving, resizing, and rotating.<br><br>The user interacts with the graphics using the mouse. Clicking on a graphic selects it, which causes the graphic to be surrounded with a selection box. The selection box has handles around it that can be dragged to resize, rotate, or move the graphic. The user can select multiple graphics by holding down the Ctrl key and clicking on each graphic. New graphics can be added to the 2D graphics list interactively using MgraInteractive().<br><br>You can restrict the type of interactions the user can have using MgraControl() or MgraControlList(). INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_DISABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_DISABLE | Specifies that the 2D graphics list associated with the display cannot be affected by user interaction. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that the 2D graphics list associated with the display can be affected by user interaction. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_GRAPHIC_LIST_OPACITY | | Sets the opacity level of annotations generated from the 2D graphics list associated with the display.<br><br>Note that this setting applies to the display, not to the associated 2D graphics list. If you associate another 2D graphics list with the display (using this function with M_ASSOCIATED_GRAPHIC_LIST_ID), the opacity that you set is also used with the new 2D graphics list. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_DISABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies not to use partial opacity. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | 0 <= *Value* <= 100 | Specifies the opacity (as a percentage), where 0 is completely transparent and 100 is completely opaque. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_INTERPOLATION_MODE | | Sets the type of interpolation used to display a zoomed image. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_NEAREST_NEIGHBOR. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_AVERAGE | Specifies averaging interpolation. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_INTERPOLATE | Specifies interpolated resizing. This gives the best speed/result compromise for interpolated resizing. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_MAX | Specifies an interpolation based on the maximum pixel value in the specified source image. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_MIN | Specifies an interpolation based on the minimum pixel value in the specified source image. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_NEAREST_NEIGHBOR | Specifies nearest neighbor interpolation. The new value is that of the pixel closest to the source point. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | c | | e | f | g | a | | i | | j | k | l | m | n | b |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_KEYBOARD_USE | Sets whether the user can interactively pan and zoom the display using the keyboard.<br><br>The default key usage is: INQ<br><br>+ — Increases the X- and Y-zoom factors.<br><br>- — Decreases the X- and Y-zoom factors.<br><br>Page-up — Scrolls the image buffer up to the previous display section.<br><br>Page-down — Scrolls the image buffer down to the next display section.<br><br>Up arrow — Scrolls the image buffer up to the previous line.<br><br>Down arrow — Scrolls the image buffer down to the next line.<br><br>Left arrow — Scrolls the image buffer left by one pixel.<br><br>Right arrow — Scrolls the image buffer right by one pixel.<br><br>Ctrl-Up arrow — Scrolls the image buffer up to the previous display section.<br><br>Ctrl-Down arrow — Scrolls the image buffer down to the next display section.<br><br>Ctrl-Left arrow — Scrolls the image buffer left to the previous display section.<br><br>Ctrl-Right arrow — Scrolls the image buffer right to the next display section. | | | | | | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_DEFAULT | For user-defined windowed displays, this value is the same as M_DISABLE; for Aurora Imaging Library windowed and exclusive displays, this value is the same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that using the keyboard to perform panning and zooming is disabled.<br><br>This is the default value for user-defined windowed displays. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that using the keyboard to perform panning and zooming is enabled.<br><br>This is the default value for Aurora Imaging Library windowed and exclusive displays. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_CURSOR_CHANGE | | Sets whether the mouse cursor changes shape in a display with mouse-interaction enabled (M_MOUSE_USE). For example, if interactive panning is possible in the display, the cursor will change to a hand.<br><br>If you have enabled M_GRAPHIC_LIST_INTERACTIVE, it is not necessary to enable this setting. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | For user-defined windowed displays, this value is the same as M_DISABLE; for Aurora Imaging Library windowed and exclusive displays, this value is the same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that mouse cursor change is disabled.<br><br>This is the default value for user-defined windowed displays. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that mouse cursor change is enabled.<br><br>This is the default value for Aurora Imaging Library windowed and exclusive displays. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_USE | | Sets whether the user can interactively pan and zoom the display using the mouse. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | For user-defined windowed displays, this value is the same as M_DISABLE; for Aurora Imaging Library windowed and exclusive displays, this value is the same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_DISABLE | Specifies that mouse-interaction will not work inside a display.<br><br>This is the default value for user-defined windowed displays. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that mouse-interaction will work inside a display.<br><br>This is the default value for Aurora Imaging Library windowed and exclusive displays. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_OVERLAY | | Sets whether to enable the overlay mechanism. This mechanism allows you to annotate displayed image buffers non-destructively. For more information on overlay buffers, see the Annotating the displayed image non-destructively section of Chapter 25: Displaying an image.<br><br>Once enabled, use MdispInquire() with the M_OVERLAY_ID inquire type to determine the identifier of the overlay buffer. Note that once created, the overlay buffer is like any other image buffer. That is, you can perform operations on it as you would on a normal buffer (except for grabbing). INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies to disable the overlay mechanism.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies to enable the overlay mechanism. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_OVERLAY_CLEAR | | Sets the value to which the overlay buffer associated with the display should be cleared. The buffer is cleared to the specified value immediately after this control type setting is changed. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_TRANSPARENT_COLOR. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_RGB888(<br>AIL_INT Red,<br>AIL_INT Green,<br>AIL_INT Blue<br>) | Specifies the RGB value to which the display's overlay buffer will be cleared for a non 8-bit display mode.<br><br>Specifies the RGB value. The buffer must be a non 8-bit 3-band buffer. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | | **Parameters** | | | | | | | | | | | | | | | |
| | | Red<br><br>Specifies the red band, as a value between 0 and 255. | | | | | | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Green<br><br>Specifies the green band, as a value between 0 and 255. | | | | | | | | | | | | | | |
| | | Blue<br><br>Specifies the blue band, as a value between 0 and 255. | | | | | | | | | | | | | | |
| | M_COLOR_BLACK | Specifies the color black. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BLUE | Specifies the color blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BRIGHT_GRAY | Specifies the color bright gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_CYAN | Specifies the color cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_BLUE | Specifies the color dark blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_CYAN | Specifies the color dark cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_GREEN | Specifies the color dark green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_MAGENTA | Specifies the color dark magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_RED | Specifies the color dark red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_YELLOW | Specifies the color dark yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GRAY | Specifies the color gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GREEN | Specifies the color green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_BLUE | Specifies the color light blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GRAY | Specifies the color light gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GREEN | Specifies the color light green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_WHITE | Specifies the color light white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_COLOR_MAGENTA | Specifies the color magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_RED | Specifies the color red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_WHITE | Specifies the color white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_YELLOW | Specifies the color yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_TRANSPARENT_COLOR | Specifies that the display's overlay buffer will be cleared to the transparency color. Set the transparency color with the M_TRANSPARENT_COLOR control type. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_OVERLAY_OPACITY | | Sets the opacity of the display's overlay buffer.<br><br>Any pixels in the overlay image which have the keying color (M_TRANSPARENT_COLOR) are always transparent, regardless of this setting. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_DISABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies not to use partial opacity. This is equivalent to setting an opacity of 100. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | 0 <= *Value* <= 100 | Specifies the opacity (as a percentage), where 0 is completely transparent and 100 is completely opaque. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_OVERLAY_SHOW | | Sets whether the display's overlay buffer is visible. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that the overlay buffer is not visible. Display the image buffer selected on the display only. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that the overlay buffer is visible. Annotations or other changes made to the overlay buffer in a color other than the transparency color will annotate the image selected to the display. Set the transparency color with the M_TRANSPARENT_COLOR control type.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_OPAQUE | Specifies that only the overlay buffer is visible. The image buffer selected will be hidden. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_REGION_INSIDE_COLOR | | Sets the color that will be used to clear the pixels that are part of a region of interest associated with the selected buffer. The pixels are cleared to the specified color immediately after this control type setting is changed. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_COLOR_GREEN. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_RGB888(<br>AIL_INT Red,<br>AIL_INT Green,<br>AIL_INT Blue<br>) | Specifies the RGB value to which the pixels that are part of a region will be cleared for a non 8-bit display mode.<br><br>Specifies the RGB value. The buffer must be a non 8-bit 3-band buffer. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | | **Parameters** | | | | | | | | | | | | | | | |
| | | Red<br><br>Specifies the red band, as a value between 0 and 255. | | | | | | | | | | | | | | | |
| | | Green<br><br>Specifies the green band, as a value between 0 and 255. | | | | | | | | | | | | | | | |
| | | Blue<br><br>Specifies the blue band, as a value between 0 and 255. | | | | | | | | | | | | | | | |
| | M_COLOR_BLACK | Specifies the color black. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BLUE | Specifies the color blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BRIGHT_GRAY | Specifies the color bright gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_CYAN | Specifies the color cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_BLUE | Specifies the color dark blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_CYAN | Specifies the color dark cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_GREEN | Specifies the color dark green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_MAGENTA | Specifies the color dark magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_COLOR_DARK_RED | Specifies the color dark red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_YELLOW | Specifies the color dark yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GRAY | Specifies the color gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GREEN | Specifies the color green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_BLUE | Specifies the color light blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GRAY | Specifies the color light gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GREEN | Specifies the color light green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_WHITE | Specifies the color light white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_MAGENTA | Specifies the color magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_RED | Specifies the color red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_WHITE | Specifies the color white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_YELLOW | Specifies the color yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_REGION_INSIDE_SHOW | | Sets how to display the pixels that are part of a region of interest associated with the selected buffer, if such a region exist. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_TRANSPARENT. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_GRAPHIC_LIST_OPACITY | Specifies that pixels in the region are displayed with a color overlay using the opacity specified with M_GRAPHIC_LIST_OPACITY. The color is specified with M_REGION_INSIDE_COLOR. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_OPAQUE | Specifies that pixels in the region are cleared to the color specified using M_REGION_INSIDE_COLOR. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_TRANSPARENT | Specifies that pixels in the region are displayed as is. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_REGION_OUTSIDE_COLOR | | Sets the color that will be used to clear the pixels that are not part of a region of interest associated with the selected buffer. The pixels are cleared to the specified color immediately after this control type setting is changed. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_COLOR_RED. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_RGB888(<br>AIL_INT Red,<br>AIL_INT Green,<br>AIL_INT Blue<br>) | Specifies the RGB value to which the pixels that are part of a region will be cleared for a non 8-bit display mode.<br><br>Specifies the RGB value. The buffer must be a non 8-bit 3-band buffer. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | | **Parameters** | | | | | | | | | | | | | | | |
| | | Red<br><br>Specifies the red band, as a value between 0 and 255. | | | | | | | | | | | | | | | |
| | | Green<br><br>Specifies the green band, as a value between 0 and 255. | | | | | | | | | | | | | | | |
| | | Blue<br><br>Specifies the blue band, as a value between 0 and 255. | | | | | | | | | | | | | | | |
| | M_COLOR_BLACK | Specifies the color black. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BLUE | Specifies the color blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BRIGHT_GRAY | Specifies the color bright gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_CYAN | Specifies the color cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_BLUE | Specifies the color dark blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_CYAN | Specifies the color dark cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_GREEN | Specifies the color dark green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_MAGENTA | Specifies the color dark magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_COLOR_DARK_RED | Specifies the color dark red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_YELLOW | Specifies the color dark yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GRAY | Specifies the color gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GREEN | Specifies the color green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_BLUE | Specifies the color light blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GRAY | Specifies the color light gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GREEN | Specifies the color light green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_WHITE | Specifies the color light white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_MAGENTA | Specifies the color magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_RED | Specifies the color red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_WHITE | Specifies the color white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_YELLOW | Specifies the color yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_REGION_OUTSIDE_SHOW | | Sets how to display the pixels that are not part of a region of interest associated with the selected buffer, if such a region exist. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_TRANSPARENT. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_GRAPHIC_LIST_OPACITY | Specifies that pixels outside of the region are displayed with a color overlay using the opacity specified with M_GRAPHIC_LIST_OPACITY. The color is specified with M_REGION_OUTSIDE_COLOR. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_OPAQUE | Specifies that pixels outside of the region are cleared to the color specified using M_REGION_OUTSIDE_COLOR. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_TRANSPARENT | Specifies that pixels outside of the region are displayed as is. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_RESTRICT_CURSOR | Sets whether to restrict the mouse cursor so that it cannot move over the exclusive display. This control type is only applicable to exclusive displays. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies to permit the mouse cursor to move over the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies to restrict the mouse cursor so that it cannot move over the display.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_SCALE_DISPLAY | Sets whether to fill the display with the selected image buffer, while keeping the same aspect ratio. You can inquire the automatically calculated zoom factor used to scale the image, using MdispInquire() with M_ZOOM_FACTOR_X and M_ZOOM_FACTOR_Y. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies that the image will not be scaled to fit the display. Unless a zoom factor has been applied, using MdispZoom(), the image will retain its original size and aspect ratio, leaving unused portions of the display to the default background color.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies to always scale the selected image buffer to fit the display, while keeping the same aspect ratio. In this case, the image is not distorted.<br><br>This setting overrides current pannning offsets (MdispPan()), as well as the previous zoom factors (MdispZoom()). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ONCE | Specifies to compute the scale factors corresponding to M_ENABLE and call MdispZoom() with these factors.<br><br>Unlike M_ENABLE, this setting changes the scale factors only once and these scale factors will not change if an image buffer with a different size is selected to the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_TITLE | Sets the display's title to the specified string. For windowed displays, set whether the window's title bar is visible using the M_WINDOW_TITLE_BAR control type. If *M_NULL* is specified, the window is given a default title ("Aurora Imaging Library Display #x"). INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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| M_NULL | Specifies to use the default title.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| AIL_TEXT("*WindowTitle*") 1 | Specifies the name of the window title (or display's title).<br><br>> Note, if using a C compiler, you must pass the value as M_PTR_TO_AIL_INT(AIL_TEXT("*WindowTitle*")). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_TRANSPARENT_COLOR | Sets the transparency (keying) color. Pixels of the image buffer selected to the display are displayed only where the corresponding pixels of the overlay buffer are set to the transparency color. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DEFAULT | Specifies the optimal transparency color. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_RGB888(<br>AIL_INT Red,<br>AIL_INT Green,<br>AIL_INT Blue<br>) | Specifies the RGB value to which to set the transparency color for a non 8-bit display mode. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | **Parameters** | | | | | | | | | | | | | | | |
| | Red<br><br>Specifies the red band, as a value between 0 and 255. | | | | | | | | | | | | | | | |
| | Green<br><br>Specifies the green band, as a value between 0 and 255. | | | | | | | | | | | | | | | |
| | Blue<br><br>Specifies the blue band, as a value between 0 and 255. | | | | | | | | | | | | | | | |
| M_COLOR_BLACK | Specifies the color black. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLOR_BLUE | Specifies the color blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLOR_BRIGHT_GRAY | Specifies the color bright gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLOR_CYAN | Specifies the color cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLOR_DARK_BLUE | Specifies the color dark blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_COLOR_DARK_CYAN | Specifies the color dark cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_GREEN | Specifies the color dark green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_MAGENTA | Specifies the color dark magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_RED | Specifies the color dark red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_YELLOW | Specifies the color dark yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GRAY | Specifies the color gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GREEN | Specifies the color green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_BLUE | Specifies the color light blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GRAY | Specifies the color light gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GREEN | Specifies the color light green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_WHITE | Specifies the color light white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_MAGENTA | Specifies the color magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_RED | Specifies the color red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_WHITE | Specifies the color white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_YELLOW | Specifies the color yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_UPDATE | | Sets whether Aurora Imaging Library should update the display. This control type can be used to temporarily disable display updates when the image buffer is being modified by more than one operation; the display can then be updated at the end of all operations. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies not to update the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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| M_ENABLE | Specifies to update the display. Also, the display is forced to update immediately.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_NOW | Specifies to force an immediate update of the display. In this case, the display is updated even if M_UPDATE is set to M_DISABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_UPDATE_GRAPHIC_LIST | Sets whether Aurora Imaging Library should update the display when modifications are made to the display's associated 2D graphics list. To associate a 2D graphics list to the display, use M_ASSOCIATED_GRAPHIC_LIST_ID.<br><br>With M_UPDATE_GRAPHIC_LIST, you can temporarily disable display updates (M_DISABLE) when the 2D graphics list is being modified by multiple operations; you can then enable display updates (M_ENABLE) at the end of all operations. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies that the display is not updated when the associated 2D graphics list is modified. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies that the display is immediately updated when the associated 2D graphics list is modified (refresh not required).<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_UPDATE_RATE_DIVIDER | Sets after how many buffer modifications to update the display, regardless of the time lapsed. For example, if you set M_UPDATE_RATE_DIVIDER to 10, one out of every 10 buffer modifications will cause a display update. If fewer than the specified number of buffer modifications occur, the display is not updated. The display is only updated with the last buffer modification. This control type is useful when performing a continuous grab.<br><br>If the display is allocated on a Distributed Aurora Imaging Library remote system but displayed on the local computer, another mechanism is also available to limit the number of display updates. You can set an upper limit to the number of display updates to perform per second, using M_UPDATE_RATE_MAX. In this case, updates of modified areas of the buffer are accumulated into one update between transmissions. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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| | M_DEFAULT | Specifies the default value; the default value is 1. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* | Specifies the frequency of the update. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_UPDATE_RATE_MAX | | Sets the maximum rate at which to update the display.<br><br>To update the display only after a specific number of buffer modifications, regardless of the time lapsed, use M_UPDATE_RATE_DIVIDER instead. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_INFINITE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_INFINITE | Specifies to perform updates as fast as possible (limited by the available bandwidth and transmission delay). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* > 0 | Specifies the maximum number of updates per second. Only integer values are accepted. A minimum delay of 1/ *Value* seconds will be respected between two consecutive updates. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_UPDATE_SYNCHRONIZATION | | [This is only applicable to Windows]<br><br>Sets the update mode currently used to update the display.<br><br>This control type allows you to serialize paint messages sent to update the display. When the display uses a user-defined window, one paint message can interrupt another. This can cause deadlock if, for example, you set up an event handler for the paint message which tries to update the image buffer selected on the display. This will cause another paint message to be called to update the display. This results in the paint message recursively calling itself, causing a deadlock. To solve this problem M_UPDATE_SYNCHRONIZATION can be set to enforce the serialization of paint messages.<br><br>This control type allows you to serialize the published image data stream with your remote display. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Specifies the default update mode.<br><br>The default mode corresponds to M_ASYNCHRONOUS, if the user-defined window thread is the same as the current one. Otherwise, it is M_SYNCHRONOUS.<br><br>Same as M_ASYNCHRONOUS. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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| | M_ASYNCHRONOUS | Specifies that the display is updated when possible.<br><br>The paint messages are queued and that the application's window procedure (WndProc) will update the display whenever possible. Basically, when a paint message is issued, Aurora Imaging Library calls the Windows `InvalidateRect()` function. For more information on `InvalidateRect()`, refer to the MSDN library.<br><br>The published image data stream to be asynchronous. The function will return immediately while internal threads are sending data to all clients that are listening for the data stream. **MdispControl**() using M_UPDATE_SYNCHRONIZATION with a valid display identifier will result in a call to the Remote module's `RemoteSetAttribute` function, with the proper parameters set. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_SYNCHRONOUS | Specifies that the display is updated immediately.<br><br>The display is updated immediately upon reception of a paint message. Basically, when a paint message is issued, Aurora Imaging Library calls the Windows `InvalidateRect()` function followed by the Windows `UpdateWindow()` function. For more information on `InvalidateRect()` and `UpdateWindow()`, refer to the MSDN library.<br><br>The published image data stream will be synchronous. The function will return only when the information is sent to all clients that are listening for the data stream. The call to **MdispControl**() using M_UPDATE_SYNCHRONIZATION with a valid display ID will result in a call to the Remote module's `RemoteSetAttribute` function, with the proper parameters set. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_VIEW_BIT_SHIFT | | Sets the number of bits by which to bit-shift the pixel values of the image buffer selected to the display, when the M_VIEW_MODE control type is set to M_BIT_SHIFT. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* | Specifies the number of bits. This value should be set to the number of significant bits in the image buffer minus 8. For example, if a 16-bit image buffer contains data grabbed from a 10-bit digitizer, a shift of 2 should be used.<br><br>The default value is 0. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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| M_VIEW_MODE | | Sets the view mode of the display. The view mode establishes how an image buffer gets remapped to the display; this is especially useful when displaying a non 8-bit image buffer. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Specifies the default value. The default value is M_TRANSPARENT unless the buffer selected on the display is a 1-bit buffer; when a 1-bit buffer is used, the default value is M_AUTO_SCALE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_AUTO_SCALE | [For essential Aurora Imaging Library Lite information, see remarks ]<br><br>Specifies to remap the pixel values to the display so that the minimum and maximum values in the image (not the full range of the image buffer) are set to 0 and 255, respectively. If the image buffer contains a single value, the displayed value is determined by linearly remapping the full range of the image buffer to that of the display (for example, (0 to 64K) to (0 to 255)). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_BIT_SHIFT | Specifies to bit-shift the pixel values of the image buffer by the specified number of bits upon updating the display. Specify the number of bits with the M_VIEW_BIT_SHIFT control type. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_MULTI_BYTES | Specifies to display each byte of the image buffer in separate display pixels. In other words, each pixel of a 16-bit image buffer will occupy two consecutive display pixels. Each pixel of a 32-bit image buffer will occupy four consecutive display pixels. This mode is only supported for 16-bit and 32-bit 1-band buffer. This mode is primarily useful when grabbing from a multi-tap camera. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_TRANSPARENT | Specifies that no pixel remapping will be performed. Note that only the 8 least-significant bits of the image buffer will be displayed. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

The following ControlType and corresponding ControlValue parameter settings are available for all windowed displays.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For windowed displays | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ControlType | | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| | ControlValue | | | | | | | | | | | | | | | | |

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| M_WINDOW_ANNOTATIONS | | Advises Aurora Imaging Library whether annotations will be drawn directly to the window's device context (DC) using GDI functions. INQ | | | | | | | | | | | | | | |
| | M_DEFAULT | [This is only applicable to Windows]<br><br>Same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_NULL | [This is only applicable to Linux]<br><br>Specifies to advise Aurora Imaging Library that the X display connection has not been set. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | [This is only applicable to Windows]<br><br>Specifies not to draw directly to the window's device context using GDI functions. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | [This is only applicable to Windows]<br><br>Specifies to draw directly to the window's device context using GDI functions. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* | [This is only applicable to Linux]<br><br>Specifies the pointer to the X display connection. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_UPDATE_ON_PAINT | | [This is only applicable to Windows]<br><br>Sets when the display will be updated. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Allows Aurora Imaging Library to decide which message to receive before updating the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Updates the display on reception of a `WM_ERASEBKGND` message in Windows. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Updates the display on reception of a `WM_PAINT` message in Windows. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

The following ControlType and corresponding ControlValue parameter settings are only available for Aurora Imaging Library windowed displays (selected using MdispSelect()), and are not available for user windowed displays (selected using MdispSelectWindow()):

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For Aurora Imaging Library windowed displays | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ControlType | | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| | ControlValue | | | | | | | | | | | | | | | | | |
| M_FULL_SCREEN | | Sets whether the window should be maximized and shown without the title bar or menu (full screen mode). INQ | | | | | | | | | | | | | | | | |
| | M_DEFAULT | Same as M_DISABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that the window will not be maximized and shown with the title bar and menu. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that the window will be maximized and shown without the title bar or menu. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_INITIAL_POSITION_X | | Sets the initial, left-most, X-coordinate of the window. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* | Specifies the X-coordinate, in pixels. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_INITIAL_POSITION_Y | | Sets the initial, top-most, Y-coordinate of the window. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* | Specifies the Y-coordinate, in pixels. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_INITIAL_SIZE_X | | Sets the initial width of the window. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Specifies that Aurora Imaging Library will calculate the optimal width. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* > 0 | Specifies the width, in pixels. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_INITIAL_SIZE_Y | | Sets the initial height of the window. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Specifies that Aurora Imaging Library will calculate the optimal height. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* > 0 | Specifies the height, in pixels. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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| M_WINDOW_MAXBUTTON | | [This is only applicable to Windows]<br><br>Sets whether the window's maximize button is visible. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that the button is not visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that the button is visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_MINBUTTON | | [This is only applicable to Windows]<br><br>Sets whether the window's minimize button is visible. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that the button is not visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that the button is visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_MOVE | | [This is only applicable to Windows]<br><br>Sets whether window movement is allowed. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies not to allow window movement. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies to allow window movement.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_OVERLAP | | [This is only applicable to Windows]<br><br>Sets whether the window can be overlapped by another. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies not to allow the window to be overlapped by another (keep window on top). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies to allow the window to be overlapped by another.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_WINDOW_RESIZE | | [This is only applicable to Windows]<br><br>Sets whether window resizing is allowed. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies not to allow window resizing. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies to allow window resizing.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_SHOW | | Sets whether the window should be shown or hidden. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that the window should be hidden. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that the window should be shown.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_SIZE_AUTO_RESET | | Sets whether the window should be resized when using MdispSelect().<br><br>By default, MdispSelect() automatically resizes the Aurora Imaging Library display window to be the same size of the selected buffer. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that Aurora Imaging Library will not resize the window. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that Aurora Imaging Library will resize the window to be the same size of the selected buffer. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_WINDOW_SYSBUTTON | | [This is only applicable to Windows]<br><br>Sets whether the window's system buttons are visible. The system buttons refer to the collection of the window's minimize, maximize, and close buttons.<br><br>To individually set whether the minimize and maximize buttons are visible, change the M_WINDOW_MAXBUTTON or M_WINDOW_MINBUTTON control type settings. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that the button is not visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that the button is visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_TITLE_BAR | | Sets whether the window's title bar is visible. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that the title bar is not visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that the title bar is visible.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_TITLE_BAR_CHANGE | | Sets whether the presence of the window's title bar can be toggled with the M_WINDOW_TITLE_BAR control type. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies not to allow toggling. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies to allow toggling.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

The following ControlType and corresponding ControlValue parameter settings are only available for user windowed displays (selected using MdispSelectWindow()), and are not available for Aurora Imaging Library windowed displays (selected using MdispSelect()):

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For user windowed displays | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ControlType | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | ControlValue | | | | | | | | | | | | | | | |
| M_GTK_MODE | | Sets whether to configure the display for use in a window created using the third-party GTK framework. Present the display in this window using MdispSelectWindow().<br><br>If this setting is disabled and the display is selected to a window created using GTK, the display might not appear correctly. INQ | | | | | | | | | | | | | | |
| | M_DISABLE | Specifies not to configure the display for use in a window created using the GTK framework.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies not to configure the display for use in a window created using the GTK framework. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_QT_MODE | | Sets whether to configure the display for use in a window created using the third-party Qt framework. Present the display in this window using MdispSelectWindow().<br><br>If this setting is disabled and the display is selected to a window created using Qt, the display might not appear correctly. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies not to configure the display for use in a window created using the Qt framework.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies to configure the display for use in a window created using the Qt framework. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_TK_MODE | | Sets whether to configure the display for use in a window created using the third-party Tk framework. Present the display in this window using MdispSelectWindow().<br><br>If this setting is disabled and the display is selected to a window created using Tk, the display might not appear correctly. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies not to configure the display for use in a window created using the Tk framework.<br><br>This is the default value. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies to configure the display for use in a window created using the Tk framework. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

The following ControlType and corresponding ControlValue parameter settings are available for displays

allocated on a Distributed Aurora Imaging Library remote system but displayed on the local computer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For displays allocated on a Distributed Aurora Imaging Library remote system but displayed on the local computer | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ControlType | | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| | ControlValue | | | | | | | | | | | | | | | | |
| M_ASYNC_UPDATE | | Sets whether to send display updates from the remote computer to the local computer in asynchronous or synchronous mode. INQ | | | | | | | | | | | | | | | |
| | M_DEFAULT | Specifies to use the default mode, which can be selected using the Aurora Imaging Configurator utility. In the Aurora Imaging Configurator utility, if you select the **High quality** display option, the default mode is M_DISABLE; if you select the **Optimized for bandwidth usage** display option, the default mode is M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies to send display updates in synchronous mode. In synchronous mode, the display is updated after each modification to the displayed image buffer (or the overlay buffer), and the application waits for the data to be transferred to the display before continuing. When an image, located on a remote computer, is displayed on the local computer, the update time can significantly slow your application. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that data will be sent asynchronously. In asynchronous mode, the required display updates are queued using multiple internal image bufferss and the application continues once the update has been queued. This mode maximizes your bandwidth usage while minimizing processing delays. There will be a delay of 1/ M_UPDATE_RATE_MAX between two consecutive updates. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_COMPRESSION_TYPE | [For essential Aurora Imaging Library Lite information, see remarks ]<br><br>Sets how to compress data being transmitted from the remote computer to the local computer. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DEFAULT | Specifies to use the default compression type, which can be selected using the Aurora Imaging Configurator utility. In the Aurora Imaging Configurator utility, if you select the **High quality** display option, compression is disabled (M_NULL); if you select the **Optimized for bandwidth usage** display option, compression is enabled and the compression type is set to M_JPEG_LOSSY. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_NULL | Specifies that compression is disabled. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_JPEG2000_LOSSLESS | Specifies that JPEG2000 lossless compression will be used. The supported data formats are: 1-band, 8- or 16-bit data. JPEG2000 compression supports much higher ratios of compression.<br><br>You might need to use this compression type, for example, when there are fine details in your image or overlay buffer. Note, however, you should avoid this compression type unless absolutely necessary since compression/decompression time is high; you should only use it when the benefits of compression outweigh the overhead associated with the compression/decompression. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_JPEG2000_LOSSY | Specifies JPEG2000 lossy compression will be used. JPEG2000 compression supports much higher ratios of compression without compromising image quality.<br><br>You might need to use this compression type, for example, when there are fine details in your image or overlay buffer. Note, however, you should avoid this compression type unless absolutely necessary since compression/decompression time is high; you should only use it when the benefits of compression outweigh the overhead associated with the compression/decompression.<br><br>You can manually set the compression factor using M_Q_FACTOR. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_JPEG_LOSSLESS | Specifies JPEG lossless compression will be used. The supported data formats are: 1-band, 8- or 16-bit data. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_JPEG_LOSSLESS_INTERLACED | Specifies that JPEG lossless compression will be used in separate fields. The supported data formats are 1-band, 8- or 16-bit data. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_JPEG_LOSSY | Specifies that JPEG lossy compression will be used.<br><br>You can manually set the compression factor using M_Q_FACTOR. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_JPEG_LOSSY_INTERLACED | Specifies that JPEG lossy compression will be used in separate fields.<br><br>You can manually set the compression factor using M_Q_FACTOR. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_Q_FACTOR | | [For essential Aurora Imaging Library Lite information, see remarks ]<br><br>Sets the quantization factor. This control type is only applied when lossy compression is used.<br><br>The Q factor is applied to all bands. INQ | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Specifies the default value; the default value is 80. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | 1 <= *Value* <= 99 | Specifies the quantization factor. Only integer values are accepted. The higher the factor, the more the compression, but the lower the image quality. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

## Remarks

- *[Aurora Imaging Library Lite]*
  Aurora Imaging Library Lite does not support displaying a 32-bit image buffer in a display that has its M_VIEW_MODE control type set to M_AUTO_SCALE, unless you have purchased the Aurora Imaging Library Image Analysis license.
- *[Aurora Imaging Library Lite]*
  Note that during development and at runtime, compression support requires the presence of an Aurora Imaging Library license that grants access to the compression/decompression package. This access is only granted by default with the development license dongle for the full version of Aurora Imaging Library. In other cases, you must purchase access to this package separately.

## Type-specific versions of the function when using a C compiler under 64-bit

```
void MdispControlDouble (AIL_ID DisplayId, AIL_INT64 ControlType, AIL_DOUBLE ControlValue)
```

Details

### Parameters

DisplayId

See DisplayId of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void MdispControlInt64 (AIL_ID DisplayId, AIL_INT64 ControlType, AIL_INT64 ControlValue)
```

Details

### Parameters

DisplayId

See DisplayId of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

## Compilation information

| | |
|---|---|
| **Header** | Include AIL.h. |
| **Library** | Use ail11.lib. |
| **DLL** | Requires ail11.dll; aildisplay11.dll. |

# MdispFree

## Synopsis

Free a display.

## Syntax

```
void MdispFree(
    AIL_ID DisplayId //in
)
```

## Description

This function deallocates a display previously allocated with MdispAlloc().

If an image buffer was selected to the display using MdispSelect(), **MdispFree**() also closes the associated window for windowed displays; if an image buffer was selected to the display using MdispSelectWindow(), the associated window is left open but it is left blank. For exclusive displays, this causes the display to disappear, allowing the desktop to reappear on the screen.

All displays allocated on a particular system must be freed before the system can be freed, unless M_UNIQUE_ID was specified during allocation.

## Parameter

DisplayId

Specifies the identifier of the display.

## Compilation information

| | |
|---|---|
| **Header** | Include AIL.h. |
| **Library** | Use ail11.lib. |
| **DLL** | Requires ail11.dll; aildisplay11.dll. |

# MdispGetHookInfo

## Synopsis

Get information about a display hook event.

## Syntax

```
AIL_INT MdispGetHookInfo(
    AIL_ID EventId,     //in
    AIL_INT64 InfoType, //in
    void *UserVarPtr    //out
)
```

## Description

This function allows you to get information about the event that caused the hook-handler function to be called. **MdispGetHookInfo**() should only be called within the scope of a display hook-handler function (see MdispHookFunction()).

## Parameters

EventId

Specifies the display event identifier received by the hook-handler function (see MdispHookFunction()).

InfoType

Specifies the type of information about the event to return.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE
- AIL_ID
- AIL_INT

Specifies the address in which to write the requested information.

See the Parameter associations section for possible values that can be returned.

## Parameter associations

The tables below list possible values for the **InfoType** parameter and possible values returned to the **UserVarPtr** parameter.

- For retrieving information about a general display hook event
- For retrieving information regarding mouse events
- For retrieving information regarding keyboard events
- For retrieving information regarding either a mouse or keyboard event

Regardless of the display event that caused the hook-handler function to be called, the InfoType parameter can be set to the value below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For retrieving information about a general display hook event | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| InfoType | | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| | UserVarPtr<br>- Possible values returned | | | | | | | | | | | | | | | | |
| M_DISPLAY | | Retrieves the Aurora Imaging Library identifier of the display that generated the event. | | | | | | | | | | | | | | | |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_ID* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Aurora Imaging Librarydisplay identifier* | Specifies the Aurora Imaging Library identifier of the display that generated the event. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

If the hook-handler function was called due to an M_MOUSE_... event type, the InfoType parameter can be set to one of the values below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For retrieving information regarding mouse events | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| InfoType | | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| | UserVarPtr<br>- Possible values returned | | | | | | | | | | | | | | | | |
| M_MOUSE_POSITION_BUFFER_X | | Retrieves the X-position of the cursor, in buffer coordinates. | | | | | | | | | | | | | | | |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_DOUBLE* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* | Specifies the X-position of the cursor, in buffer coordinates. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_POSITION_BUFFER_Y | | Retrieves the Y-position of the cursor, in buffer coordinates. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_DOUBLE* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* | Specifies the Y-position of the cursor, in buffer coordinates. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_POSITION_X | | Retrieves the X-position of the cursor, in display coordinates. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* | Specifies the X-position of the cursor, in display coordinates. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_POSITION_Y | Retrieves the Y-position of the cursor, in display coordinates. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* | Specifies the Y-position of the cursor, in display coordinates. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_WHEEL_VALUE | Retrieves the value for the mouse wheel's rotation. Note, this value can only be retrieved if the hook-handler function was called due to the M_MOUSE_WHEEL event. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* | Specifies the value for the mouse wheel's rotation. A positive value indicates that the wheel was rotated forward, away from the user; a negative value indicates that the wheel was rotate backward, toward the user. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

If the hook-handler function was called due to an M_KEY_... event type, the InfoType parameter can be set to the value below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For retrieving information regarding keyboard events | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| InfoType | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| UserVarPtr - Possible values returned | | | | | | | | | | | | | | | | |
| M_AIL_KEY_VALUE | Retrieves the Aurora Imaging Library operating system-independent key code for the keyboard key that triggered the event. Note, this value can only be retrieved if the hook-handler function was called due to the M_KEY_DOWN or M_KEY_UP event. | | | | | | | | | | | | | | | |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_KEY_ADD | Specifies the Add key (+). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_ARROW_DOWN | Specifies the Down Arrow key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_ARROW_LEFT | Specifies the Left Arrow key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_ARROW_RIGHT | Specifies the Right Arrow key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_ARROW_UP | Specifies the Up Arrow key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_BACK | Specifies the Back key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_CLEAR | Specifies the Clear key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_DECIMAL | Specifies the Period key located on your number pad. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_DELETE | Specifies the Delete key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_DIVIDE | Specifies the Divide key (/). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_END | Specifies the End key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_ESC | Specifies the Esc key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_EXECUTE | Specifies the Execute key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_F*n* | Specifies the *n* key, where *n* is a value from 1 to 24. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_HOME | Specifies the Home key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_INSERT | Specifies the Insert key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_MULTIPLY | Specifies the Multiply key (*). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_*n* | Specifies the *n* key, where *n* is a value from 0 to 9. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_NUM_LOCK | Specifies the Num Lock key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_KEY_NUMPAD*n* | Specifies the *n* key, where *n* is a value from 0 to 9. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_PAGEDOWN | Specifies the Pagedown key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_PAGEUP | Specifies the Pageup key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_PAUSE | Specifies the Pause key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_RETURN | Specifies the Return key (Enter). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_SCROLL_LOCK | Specifies the Scroll Lock key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_SEPARATOR | Specifies the Separator key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_SPACE | Specifies the Space key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_SUBTRACT | Specifies the Subtract key (-). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_TAB | Specifies the Tab key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_*xxx* | Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_KEY_VALUE | | Retrieves the operating system's key code for the keyboard key that triggered the event. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* | Specifies the key code. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

If the hook-handler function was called due to either an M_KEY_... or M_MOUSE_... event type, the InfoType parameter can be set to the value below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For retrieving information regarding either a mouse or keyboard event | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| InfoType | | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| | UserVarPtr<br>- Possible values returned | | | | | | | | | | | | | | | | |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_COMBINATION_KEYS | | Retrieves which combination keys or mouse buttons triggered the event. Note that a combination of the values below can be returned. Bitwise operators must be used to establish the specific keys or buttons pressed. | | | | | | | | | | | | | | | |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_ALT | Specifies the Alt key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_CTRL | Specifies the Ctrl key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_SHIFT | Specifies the Shift key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_KEY_WIN | Specifies the Windows key. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_MOUSE_LEFT_BUTTON | Specifies the left mouse button. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_MOUSE_MIDDLE_BUTTON | Specifies the middle mouse button. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_MOUSE_RIGHT_BUTTON | Specifies the right mouse button. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

## Return value

The returned value is **M_NULL** if successful. If the operation fails, a non-null (!**M_NULL**) value is returned.

## Compilation information

| | |
|---|---|
| **Header** | Include AIL.h. |
| **Library** | Use ail11.lib. |
| **DLL** | Requires ail11.dll; aildisplay11.dll. |

# MdispHookFunction

## Synopsis

Hook a function to a display event.

## Syntax

```
void MdispHookFunction(
    AIL_ID DisplayId,                          //out
    AIL_INT HookType,                          //in
    AIL_DISP_HOOK_FUNCTION_PTR HookHandlerPtr, //in
    void *UserDataPtr                          //in-out
)
```

## Description

This function allows you to attach or detach a user-defined function to a specified display event. Once a hook-handler function is defined and hooked to an event, it is automatically called when the event occurs.

You can hook more than one function to an event by making separate calls to **MdispHookFunction**() for each function that you want to hook. Aurora Imaging Library automatically chains and keeps an internal list of all these hooked functions. When a function is hooked, this new function is added to the end of the list. When the event happens, all user-defined functions in the list will be executed in the same order that they were hooked to the event. You can also remove any function from the list; in this case, Aurora Imaging Library preserves the order of the remaining functions in the list.

Note that this function is not available on network displays.

## Parameters

DisplayId

Specifies the identifier of the target display.

HookType

Specifies the display event type. This parameter can be set to the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For specifying the display event type | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| M_KEY_CHAR + | Calls the hook-handler function each time the user presses a character key, while the mouse cursor is in the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_KEY_DOWN + | Calls the hook-handler function each time the user presses a keyboard key, while the mouse cursor is in the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_KEY_UP + | Calls the hook-handler function each time the user releases a keyboard key, while the mouse cursor is in the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_LEFT_BUTTON_DOWN + | Calls the hook-handler function each time the user clicks the left mouse button, while the mouse cursor is in the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_LEFT_BUTTON_UP + | Calls the hook-handler function each time the user releases the left mouse button, while the mouse cursor is in the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_LEFT_DOUBLE_CLICK + | Calls the hook-handler function each time the user double-clicks the left mouse button, while the mouse cursor is in the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_MIDDLE_BUTTON_DOWN + | Calls the hook-handler function each time the user clicks the middle mouse button, while the mouse cursor is in the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_MIDDLE_BUTTON_UP + | Calls the hook-handler function each time the user releases the middle mouse button, while the mouse cursor is in the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_MOVE + | Calls the hook-handler function each time the mouse cursor moves, while in the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_RIGHT_BUTTON_DOWN + | Calls the hook-handler function each time the user clicks the right mouse button, while the mouse cursor is in the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_RIGHT_BUTTON_UP + | Calls the hook-handler function each time the user releases the right mouse button, while the mouse cursor is in the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_WHEEL + | Calls the hook-handler function each time the user scrolls the mouse wheel (up or down), while the mouse cursor is in the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

Combination value for the values listed in For specifying the display event type.

You can add the following value to the above-mentioned values to set whether to detach the hook-handler function.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For detaching the hook-handler function | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Combination value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| M_UNHOOK | Unhooks the specified function if hooked to the specified event. When you use M_UNHOOK, you must provide the same values for all the parameters as when you originally hooked the function. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

HookHandlerPtr

Specifies the address of the function that should be called when the specified event occurs. The hook-handler function, pointed to by HookHandlerPtr, must be declared as follows:

```
AIL_INT MFTYPE HookHandler(
    AIL_INT HookType,
    AIL_ID EventId,
    void *UserDataPtr
)
```

Parameters

HookType

Type of event hooked.

EventId

Event identifier to pass to MdispGetHookInfo() when inquiring about the hooked event.

UserDataPtr

Data pointer that was passed to **MdispHookFunction**().

Upon successful completion, the hook-handler function should return **M_NULL**. Note, MFTYPE and AIL_DISP_HOOK_FUNCTION_PTR are reserved Aurora Imaging Library predefined types for functions and data pointers.

UserDataPtr

Specifies the address of the user data that you want to make available to the hook-handler function. This address is passed to the hook-handler function, through its UserDataPtr parameter, when the specified event occurs. Set this parameter to **M_NULL** if not used.

## Compilation information

| | |
|---|---|
| **Header** | Include AIL.h. |
| **Library** | Use ail11.lib. |

| | |
|---|---|
| **DLL** | Requires ail11.dll; aildisplay11.dll. |

# MdispInquire

## Synopsis

Inquire about a display setting.

## Syntax

```
AIL_INT MdispInquire(
    AIL_ID DisplayId,      //in
    AIL_INT64 InquireType, //in
    void *UserVarPtr       //out
)
```

## Description

This function inquires about a specified display setting.

## Parameters

DisplayId

Specifies the identifier of the display.

InquireType

Specifies the type of display setting about which to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_TEXT_CHAR [optionally, in C++: a reference to an AIL_STRING]
- AIL_DOUBLE
- AIL_ID
- AIL_INT
- AIL_INT64
- AIL_WINDOW_HANDLE

Specifies the address in which to write the requested information. Since the **MdispInquire**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

## Parameter associations

The tables below list possible values for the **InquireType** parameter and possible values returned to the **UserVarPtr** parameter.

- For all types of displays
- For all windowed displays
- For Aurora Imaging Library windowed displays
- For user windowed displays
- For displays allocated on a Distributed Aurora Imaging Library remote system but displayed on the

local computer

Unless otherwise specified, the following InquireType parameter settings are available for all types of displays.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For all types of displays | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| InquireType | | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| | UserVarPtr<br>- Possible values returned | | | | | | | | | | | | | | | | |
| M_ASSOCIATED_GRAPHIC_LIST_ID | | Inquires the identifier of the 2D graphics list. SET | | | | | | | | | | | | | | | |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_ID* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_NULL | Specifies that no 2D graphics list is associated with the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *2D graphics list identifier* | Specifies the identifier of the 2D graphics list that is associated with the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_BACKGROUND_COLOR | | Inquires the background color of the display. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BLACK | Specifies the color black. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BLUE | Specifies the color blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BRIGHT_GRAY | Specifies the color bright gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_CYAN | Specifies the color cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_BLUE | Specifies the color dark blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_CYAN | Specifies the color dark cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_GREEN | Specifies the color dark green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_MAGENTA | Specifies the color dark magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_COLOR_DARK_RED | Specifies the color dark red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_YELLOW | Specifies the color dark yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GRAY | Specifies the color gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GREEN | Specifies the color green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_BLUE | Specifies the color light blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GRAY | Specifies the color light gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GREEN | Specifies the color light green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_WHITE | Specifies the color light white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_MAGENTA | Specifies the color magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_RED | Specifies the color red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_WHITE | Specifies the color white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_YELLOW | Specifies the color yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Byte-encoded RGB value* | Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_CENTER_DISPLAY | | Inquires whether a selected image buffer should be centered in the display. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that the image buffer will not be centered in the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_ENABLE | Specifies that the image buffer will always be centered in the display, both in X and Y. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISPLAY_TYPE | | Inquires the display type. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_EXCLUSIVE + | Specifies that the display is an exclusive display.<br>> This display type requires MdispSelect() to display images. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_WEB | Specifies that the display is an Aurora Imaging Web Library display.<br>> This display type requires MdispSelect() to display images. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_WINDOWED + | Specifies that the display is a windowed display.<br>> This display type can be used with MdispSelect() and MdispSelectWindow(). A remote display requires MdispSelect() to display images. To inquire if a display is remote, use M_EXTENDED_INIT_FLAG. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_WPF | Specifies that the display is a WPF display.<br>> This display type requires MdispSelect() to display images. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_EXTENDED_INIT_FLAG | | Inquires the initialization flag specified upon allocating the display. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT64* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_EXCLUSIVE + | Specifies to present the image buffer selected for display full-screen, without a windowed border, in one of Windows desktop screens. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_WEB | Specifies to present the image buffer selected for display in one or more instances of an Aurora Imaging Web Library client application (for example, running in a compatible web browser). INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_WINDOWED + | Specifies to present the image buffer selected for display in its own window on the Windows desktop screen(s). INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_WPF | Specifies to present the image buffer selected for display using an AILWPFDisplay control. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_FORMAT + | | Inquires the display's data format. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING ]* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_CURRENT_RESOLUTION | //For exclusive displays.<br>Specifies to use the current resolution of the screen. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | AIL_TEXT("M_DEFAULT") 1 | //For specifying the default display format.<br>Specifies the default display format. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | AIL_TEXT("*vcf-file-name.vcf*") 1 | //For exclusive displays.<br>Specifies the name of the video configuration format (VCF) that defines the resolution and refresh rate to use. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_GRAPHIC_LIST_INTERACTIVE | | Inquires whether a user can interactively modify the graphics in the 2D graphics list associated with the display. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Same as M_DISABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that the 2D graphics list associated with the display cannot be affected by user interaction. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_ENABLE | Specifies that the 2D graphics list associated with the display can be affected by user interaction. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_GRAPHIC_LIST_OPACITY | Inquires the opacity of annotations generated from the 2D graphics list associated with the display. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies not to use partial opacity. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| 0 <= *Value* <= 100 | Specifies the opacity (as a percentage), where 0 is completely transparent and 100 is completely opaque. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_INTERPOLATION_MODE | Inquires the type of interpolation used to display a zoomed image. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_AVERAGE | Specifies averaging interpolation. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_INTERPOLATE | Specifies interpolated resizing. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MAX | Specifies an interpolation based on the maximum pixel value in the specified source image. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MIN | Specifies an interpolation based on the minimum pixel value in the specified source image. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_NEAREST_NEIGHBOR | Specifies nearest neighbor interpolation. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_KEYBOARD_USE | Inquires whether the user can interactively pan and zoom the display using the keyboard. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_DISABLE | Specifies that using the keyboard to perform panning and zooming is disabled. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that using the keyboard to perform panning and zooming is enabled. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_LUT_ID | | Inquires the identifier of the LUT buffer associated with the display. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_ID* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Specifies that any association to a LUT buffer is removed from the specified display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLORMAP_DISTINCT_256 | Specifies the LUT buffer with a distinct colormap as the LUT buffer that is associated with the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLORMAP_GRAYSCALE | Specifies the LUT buffer with a grayscale colormap as the LUT buffer that is associated with the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLORMAP_HOT | Specifies the LUT buffer with a hot colormap as the LUT buffer that is associated with the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLORMAP_HUE | Specifies the LUT buffer with a hue colormap as the LUT buffer that is associated with the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLORMAP_JET | Specifies the LUT buffer with a jet colormap as the LUT buffer that is associated with the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLORMAP_SPECTRUM | Specifies the LUT buffer with a spectrum colormap as the LUT buffer that is associated with the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLORMAP_TURBO | Specifies the LUT buffer with a turbo colormap as the LUT buffer that is associated with the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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| M_COLORMAP_WHEEL | Specifies the LUT buffer with a colorwheel colormap as the LUT buffer that is associated with the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_PSEUDO | Specifies the default pseudo-color LUT buffer as the LUT buffer that is associated with the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *LUT buffer identifier* | Specifies the identifier of the custom LUT buffer (allocated with MbufAlloc1d() or MbufAllocColor() with M_LUT) that is associated with the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_LUT_SUPPORTED | Inquires whether LUTs are supported by the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_NO | Specifies that LUTs are not supported. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_YES | Specifies that LUTs are supported. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_CURSOR_CHANGE | Inquires whether the mouse cursor changes shape in a display with mouse-interaction enabled. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DEFAULT | For user-defined windowed displays, this value is the same as M_DISABLE; for Aurora Imaging Library windowed and exclusive displays, this value is the same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies that mouse cursor change is disabled. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies that mouse cursor change is enabled. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MOUSE_USE | Inquires whether the user can interactively pan and zoom the display using the mouse. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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| M_DEFAULT | For user-defined windowed displays, this value is the same as M_DISABLE; for Aurora Imaging Library windowed and exclusive displays, this value is the same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies that mouse-interaction will not work inside a display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies that mouse-interaction will work inside a display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_NUMBER | Inquires the display device's rank in the system. Note that the returned value can also be a combination of M_TOP, M_BOTTOM, or M_CENTER + M_LEFT or M_RIGHT, specifying the position of the display device in screen arrangements of up to 3x3 screens. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DEFAULT | Specifies that Aurora Imaging Library will use the most appropriate device for display purposes. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_BOTTOM | Specifies to use the bottom-most device for an exclusive display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_BOTTOM + M_LEFT | Specifies to use the bottom left device for an exclusive display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_BOTTOM + M_RIGHT | Specifies to use the bottom right device for an exclusive display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_CENTER | Specifies to use the center device for an exclusive display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_CENTER + M_LEFT | Specifies to use the center left device for an exclusive display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_CENTER + M_RIGHT | Specifies to use the center right device for an exclusive display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_LEFT | Specifies to use the left-most device for an exclusive display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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| | M_RIGHT | Specifies to use the right-most device for an exclusive display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_TOP | Specifies to use the top-most device for an exclusive display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_TOP + M_LEFT | Specifies to use the top left device for an exclusive display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_TOP + M_RIGHT | Specifies to use the top right device for an exclusive display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_OVERLAY | | Inquires whether or not the overlay mechanism is enabled for the display. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies to disable the overlay mechanism. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies to enable the overlay mechanism. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_OVERLAY_ID | | Inquires the identifier of the overlay buffer associated with the display.<br><br>Note that the overlay buffer is a normal buffer and, knowing its Aurora Imaging Library identifier, you can perform operations on it as you would on a normal buffer (except for grabbing). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_ID* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_NULL | Specifies that there is no overlay buffer associated with the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Buffer identifier* | Specifies the Aurora Imaging Library identifier of the overlay buffer. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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| M_OVERLAY_OPACITY | Inquires the opacity of the display's overlay buffer.<br><br>Any pixels in the overlay image which have the keying color (M_TRANSPARENT_COLOR) are always transparent, regardless of this setting. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies not to use partial opacity. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| 0 <= *Value* <= 100 | Specifies the opacity (as a percentage), where 0 is completely transparent and 100 is completely opaque. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_OVERLAY_SHOW | Inquires the visible state of the overlay. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies that the overlay buffer is not visible. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies that the overlay buffer is visible. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_OPAQUE | Specifies that only the overlay buffer is visible. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_OWNER_SYSTEM | Inquires the identifier of the system on which the display has been allocated. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_ID* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DEFAULT_HOST | Specifies the default Host system of the current Aurora Imaging Library application. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *System identifier* | Specifies a valid system identifier, previously allocated using MsysAlloc(). | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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| M_PAN_OFFSET_X | Inquires the number of pixels in X by which the image buffer is panned.<br><br>Note that when you interactively pan the image with the mouse or keyboard, M_PAN_OFFSET_X will reflect the offset of the image. SET SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_DOUBLE* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* | Specifies the number of pixels by which the image is panned horizontally. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_PAN_OFFSET_Y | Inquires the number of pixels in Y by which to the image buffer is panned.<br><br>Note that when you interactively pan the image with the mouse or keyboard, M_PAN_OFFSET_Y will reflect the offset of the image. SET SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_DOUBLE* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* | Specifies the number of pixels by which the image is panned vertically. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_PIXEL_FORMAT | Inquires the pixel format of the current display resolution set by your graphics controller. Allocating a displayable image buffer with the same format will ensure maximum performance with regards to display updates. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_BGR24 | Specifies 24-bit BGR format. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_BGR32 | Specifies 32-bit BGR format. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MONO8 | Specifies 8-bit monochrome format. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_RGB15 | Specifies 15-bit RGB format. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_RGB16 | Specifies 16-bit RGB format. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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| | M_YUV16_UYVY | Specifies YUV16 format, stored in the UYVY order. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_YUV16_YUYV | Specifies YUV16 format, stored in the YUYV order. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_REGION_INSIDE_COLOR | | Inquires the color that will be used to clear the pixels that are part of a region of interest associated with the selected buffer. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BLACK | Specifies the color black. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BLUE | Specifies the color blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BRIGHT_GRAY | Specifies the color bright gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_CYAN | Specifies the color cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_BLUE | Specifies the color dark blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_CYAN | Specifies the color dark cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_GREEN | Specifies the color dark green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_MAGENTA | Specifies the color dark magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_RED | Specifies the color dark red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_YELLOW | Specifies the color dark yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GRAY | Specifies the color gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GREEN | Specifies the color green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_BLUE | Specifies the color light blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GRAY | Specifies the color light gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_COLOR_LIGHT_GREEN | Specifies the color light green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLOR_LIGHT_WHITE | Specifies the color light white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLOR_MAGENTA | Specifies the color magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLOR_RED | Specifies the color red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLOR_WHITE | Specifies the color white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLOR_YELLOW | Specifies the color yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Byte-encoded RGB value* | Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_REGION_INSIDE_SHOW | Inquires how to display the pixels that are part of a region of interest associated with the selected buffer, if such a region exist. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_GRAPHIC_LIST_OPACITY | Specifies that pixels in the region are displayed with a color overlay using the opacity specified with M_GRAPHIC_LIST_OPACITY. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_OPAQUE | Specifies that pixels in the region are cleared to the color specified using M_REGION_INSIDE_COLOR. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_TRANSPARENT | Specifies that pixels in the region are displayed as is. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_REGION_OUTSIDE_COLOR | Inquires the color that will be used to clear the pixels that are not part of a region of interest associated with the selected buffer. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_COLOR_BLACK | Specifies the color black. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BLUE | Specifies the color blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_BRIGHT_GRAY | Specifies the color bright gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_CYAN | Specifies the color cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_BLUE | Specifies the color dark blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_CYAN | Specifies the color dark cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_GREEN | Specifies the color dark green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_MAGENTA | Specifies the color dark magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_RED | Specifies the color dark red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_YELLOW | Specifies the color dark yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GRAY | Specifies the color gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GREEN | Specifies the color green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_BLUE | Specifies the color light blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GRAY | Specifies the color light gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GREEN | Specifies the color light green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_WHITE | Specifies the color light white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_MAGENTA | Specifies the color magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_RED | Specifies the color red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_WHITE | Specifies the color white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_YELLOW | Specifies the color yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Byte-encoded RGB value* | Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_REGION_OUTSIDE_SHOW | Inquires how to display the pixels that are not part of a region of interest associated with the selected buffer, if such a region exist. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_GRAPHIC_LIST_OPACITY | Specifies that pixels outside of the region are displayed with a color overlay using the opacity specified with M_GRAPHIC_LIST_OPACITY. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_OPAQUE | Specifies that pixels outside of the region are cleared to the color specified using M_REGION_OUTSIDE_COLOR. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_TRANSPARENT | Specifies that pixels outside of the region are displayed as is. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_RESTRICT_CURSOR | Inquires whether the mouse cursor is restricted so that it cannot move over the exclusive display. This inquire type is only applicable to exclusive displays. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies to permit the mouse cursor to move over the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies to restrict the mouse cursor so that it cannot move over the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_SCALE_DISPLAY | Inquires whether a selected buffer will fill the display, maintaining its aspect ratio. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_DISABLE | Specifies that the image will not be scaled to fit the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies to always scale the selected image buffer to fit the display, while keeping the same aspect ratio. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_SELECTED | Inquires the identifier of the image buffer currently displayed. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_ID* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_NULL | Specifies that no buffer is currently being displayed. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Buffer identifier* | Specifies the Aurora Imaging Library identifier of the image buffer selected to the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_SIZE_BAND | Inquires the number of color bands that the display is capable of displaying. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* | Specifies the number of color bands. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_SIZE_BAND_LUT | Inquires the number of color bands of the LUT (if any) associated with the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* | Specifies the number of color bands. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_SIZE_BIT | Inquires the number of bits (depth) of the display per band. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* | Specifies the depth per band, in bits. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_SIZE_X | Inquires the display's width. For windowed displays, the returned value is equal to the width of the client area of the window. For exclusive displays, the returned width is the width specified in the VCF file. For Aurora Imaging Web Library displays, the returned value is equal to the width (M_SIZE_X) of the image buffer currently selected to the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* | Specifies the display's width, in pixels. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_SIZE_Y | Inquires the display's height. For windowed displays, the returned value is equal to the height of the client area of the window. For exclusive displays, the returned height is the height specified in the VCF file. For Aurora Imaging Web Library displays, the returned value is equal to the height (M_SIZE_Y) of the image buffer currently selected to the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* | Specifies the display's height, in pixels. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_TITLE + | Inquires the display's title. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING ]* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| AIL_TEXT("*Title*") 1 | Specifies the title of the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_TRANSPARENT_COLOR | Inquires the transparency (keying) color. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLOR_BLACK | Specifies the color black. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLOR_BLUE | Specifies the color blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_COLOR_BRIGHT_GRAY | Specifies the color bright gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_CYAN | Specifies the color cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_BLUE | Specifies the color dark blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_CYAN | Specifies the color dark cyan. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_GREEN | Specifies the color dark green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_MAGENTA | Specifies the color dark magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_RED | Specifies the color dark red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_DARK_YELLOW | Specifies the color dark yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GRAY | Specifies the color gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_GREEN | Specifies the color green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_BLUE | Specifies the color light blue. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GRAY | Specifies the color light gray. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_GREEN | Specifies the color light green. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_LIGHT_WHITE | Specifies the color light white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_MAGENTA | Specifies the color magenta. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_RED | Specifies the color red. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_WHITE | Specifies the color white. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_COLOR_YELLOW | Specifies the color yellow. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Byte-encoded RGB value* | Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_UPDATE | Inquires whether Aurora Imaging Library should update the display. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies not to update the display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies to update the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_UPDATE_GRAPHIC_LIST | Inquires whether Aurora Imaging Library should update the display when modifications are made to the display's associated 2D graphics list. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies that the display is not updated when the associated 2D graphics list is modified. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies that the display is immediately updated when the associated 2D graphics list is modified (refresh not required). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_UPDATE_RATE | Inquires the display update rate. This is the number of display updates that occurred during the last second. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_DOUBLE* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| 0 | Specifies that the buffer selected to the display was not modified. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* > 0 | Specifies the display update rate. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_UPDATE_RATE_DIVIDER | Inquires how often the display should be updated with buffer modifications. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* | Specifies the frequency of the update. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_UPDATE_RATE_MAX | Inquires the maximum rate at which to perform updates. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_INFINITE | Specifies to perform updates as fast as possible (limited by the available bandwidth and transmission delay). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* > 0 | Specifies the maximum number of updates per second. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_UPDATE_SYNCHRONIZATION | [This is only applicable to Windows]<br>Inquires the update mode currently used by the display for Windows paint messages. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DEFAULT | [This is only applicable to Windows]<br>Specifies the default update mode. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ASYNCHRONOUS | [This is only applicable to Windows]<br>Specifies that the display is updated when possible. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_SYNCHRONOUS | [This is only applicable to Windows]<br>Specifies that the display is updated immediately. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_VIEW_BIT_SHIFT | Inquires the number of bits by which the buffer data gets shifted when M_VIEW_MODE is set to M_BIT_SHIFT. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Value* | Specifies the number of bits. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_VIEW_MODE | Inquires how a buffer gets remapped to the display (view mode). SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DEFAULT | Specifies the default value. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_AUTO_SCALE | Specifies to remap the pixel values to the display so that the minimum and maximum values in the image (not the full range of the image buffer) are set to 0 and 255, respectively. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_BIT_SHIFT | Specifies to bit-shift the pixel values of the image buffer by the specified number of bits upon updating the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_MULTI_BYTES | Specifies to display each byte of the image buffer in separate display pixels. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_TRANSPARENT | Specifies that no pixel remapping will be performed. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ZOOM_FACTOR_X | Inquires the zoom factor in the X-direction.<br><br>Note that when you interactively zoom the image with the mouse or keyboard, M_ZOOM_FACTOR_X will reflect the zoom of the image. SET SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_DOUBLE* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *Value* > 0.0 | Specifies the zoom factor for the X-direction of the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ZOOM_FACTOR_Y | Inquires the zoom factor in the Y-direction.<br><br>Note that when you interactively zoom the image with the mouse or keyboard, M_ZOOM_FACTOR_Y will reflect the zoom of the image. SET SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_DOUBLE* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* > 0.0 | Specifies the zoom factor for the Y-direction of the display. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |

[1] If the value was passed in a variable, it won't be returned enclosed in AIL_TEXT().

Combination value for [when InquireType=M_EXTENDED_INIT_FLAG: M_EXCLUSIVE; M_WINDOWED].

The following value might be combined with the above-mentioned values to determine whether the display was allocated on a remote computer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| Returns whether the display was allocated on a remote computer | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Combination value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| M_REMOTE_DISPLAY | Specifies that the display is displayed on the remote computer in a Distributed Aurora Imaging Library application. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |

Combination value for M_FORMAT; M_TITLE.

You can add the following value to the above-mentioned values to get the string's length.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For getting the string size | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| InquireType | | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| | UserVarPtr - Possible values returned | | | | | | | | | | | | | | | | |
| M_STRING_SIZE | | Retrieves the length of the string, including the terminating null character ("\0"). | | | | | | | | | | | | | | | |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |

Combination values for [when InquireType=M_DISPLAY_TYPE: M_EXCLUSIVE; M_WINDOWED].

One of the following values might be combined with the above-mentioned values to specify whether the display is an Aurora Imaging Library window or a user-defined window.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| Returns whether the display is an Aurora Imaging Library window or a user-defined window | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Combination value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_AIL_WINDOW | Specifies that the display is an Aurora Imaging Library display, selected using MdispSelect(). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_USER_WINDOW | Specifies that the display is a user-defined display, selected using MdispSelectWindow(). | c | | e | f | g | a | | i | | j | k | l | m | n | b |

The following InquireType parameter settings are available for all windowed displays.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For all windowed displays | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| InquireType | | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| | UserVarPtr<br>- Possible values returned | | | | | | | | | | | | | | | | |
| M_WINDOW_ANNOTATIONS | | Inquires whether annotations will be drawn directly to the window's device context (DC) using GDI functions. SET | | | | | | | | | | | | | | | |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_NULL | [This is only applicable to Linux]<br><br>Specifies to advise Aurora Imaging Library that the X display connection has not been set. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | [This is only applicable to Windows]<br><br>Specifies not to draw directly to the window's device context using GDI functions. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | [This is only applicable to Windows]<br><br>Specifies to draw directly to the window's device context using GDI functions. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* | [This is only applicable to Linux]<br><br>Specifies the pointer to the X display connection. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_UPDATE_ON_PAINT | | [This is only applicable to Windows]<br><br>Inquires when the display will be updated. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_DISABLE | [This is only applicable to Windows]<br><br>Updates the display on reception of a `WM_ERASEBKGND` message in Windows. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | [This is only applicable to Windows]<br><br>Updates the display on reception of a `WM_PAINT` message in Windows. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

The following are only available for Aurora Imaging Library windowed displays (selected using MdispSelect()) and are not available for user windowed displays (selected using MdispSelectWindow()):

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For Aurora Imaging Library windowed displays | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| InquireType | | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| | UserVarPtr<br>- Possible values returned | | | | | | | | | | | | | | | | |
| M_FULL_SCREEN | | Inquires whether the window should be maximized and shown without the title bar or menu (full screen mode).<br>SET | | | | | | | | | | | | | | | |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies that the window will not be maximized and shown with the title bar and menu. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that the window will be maximized and shown without the title bar or menu. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_HANDLE | | Inquires the Windows handle (*HWND*) of the display's window. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_WINDOW_HANDLE* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* | Specifies the Windows handle of the display's window. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_INITIAL_POSITION_X | | Inquires the initial, left-most, X-coordinate specified for the window.<br>SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | *Value* | Specifies the X-coordinate, in pixels. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_INITIAL_POSITION_Y | | Inquires the initial, left-most, Y-coordinate specified for the window. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* | Specifies the Y-coordinate, in pixels. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_INITIAL_SIZE_X | | Inquires the initial width of the window. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Specifies that Aurora Imaging Library will calculate the optimal width. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* > 0 | Specifies the width, in pixels. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_INITIAL_SIZE_Y | | Inquires the initial height of the window. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Specifies that Aurora Imaging Library will calculate the optimal height. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *Value* > 0 | Specifies the height, in pixels. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_MAXBUTTON | | [This is only applicable to Windows]<br><br>Inquires whether the window's maximize button should be visible. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | [This is only applicable to Windows]<br><br>Same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | [This is only applicable to Windows]<br><br>Specifies that the button is not visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_ENABLE | [This is only applicable to Windows]<br><br>Specifies that the button is visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_MINBUTTON | | [This is only applicable to Windows]<br><br>Inquires whether the window's minimize button should be visible. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | [This is only applicable to Windows]<br><br>Same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | [This is only applicable to Windows]<br><br>Specifies that the button is not visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | [This is only applicable to Windows]<br><br>Specifies that the button is visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_MOVE | | [This is only applicable to Windows]<br><br>Inquires whether window movement is allowed. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | [This is only applicable to Windows]<br><br>Specifies not to allow window movement. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | [This is only applicable to Windows]<br><br>Specifies to allow window movement. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_OVERLAP | | [This is only applicable to Windows]<br><br>Inquires whether the window can be overlapped by another. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | [This is only applicable to Windows]<br><br>Specifies not to allow the window to be overlapped by another (keep window on top). | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | [This is only applicable to Windows]<br><br>Specifies to allow the window to be overlapped by another. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_WINDOW_RESIZE | [This is only applicable to Windows]<br><br>Inquires whether window resizing is allowed. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | [This is only applicable to Windows]<br><br>Specifies not to allow window resizing. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | [This is only applicable to Windows]<br><br>Specifies to allow window resizing. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_SHOW | Inquires whether to show or hide the window. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies that the window should be hidden. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies that the window should be shown. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_SIZE_AUTO_RESET | Inquires whether the window should be resized when using MdispSelect().<br><br>By default, MdispSelect() automatically resizes the Aurora Imaging Library display window to be the same size of the selected buffer. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies that Aurora Imaging Library will not resize the window. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies that Aurora Imaging Library will resize the window to be the same size of the selected buffer. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_SYSBUTTON | [This is only applicable to Windows]<br><br>Inquires whether the window system buttons (minimize, maximize, and close buttons) should be visible. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_DEFAULT | [This is only applicable to Windows]<br>Same as M_ENABLE. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | [This is only applicable to Windows]<br>Specifies that the button is not visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | [This is only applicable to Windows]<br>Specifies that the button is visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_TITLE_BAR | Inquires whether the title bar should be visible. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies that the title bar is not visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies that the title bar is visible. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_WINDOW_TITLE_BAR_CHANGE | Inquires whether the presence of the window's title bar can be toggled with the M_WINDOW_TITLE_BAR control type. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies not to allow toggling. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies to allow toggling. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

The following are only available for user windowed displays (selected using MdispSelectWindow()) and are not available for Aurora Imaging Library windowed displays (selected using MdispSelect()):

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For user windowed displays | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| InquireType | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| UserVarPtr<br>- Possible values returned | | | | | | | | | | | | | | | | |
| M_GTK_MODE | Inquires whether the display is configured for use in a window created using the third-party GTK framework. SET | | | | | | | | | | | | | | | |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_DISABLE | Specifies not to configure the display for use in a window created using the GTK framework. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies not to configure the display for use in a window created using the GTK framework. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_QT_MODE | Inquires whether the display is configured for use in a window created using the third-party Qt framework. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies not to configure the display for use in a window created using the Qt framework. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies to configure the display for use in a window created using the Qt framework. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_TK_MODE | Inquires whether the display is configured for use in a window created using the third-party Tk framework. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_DISABLE | Specifies not to configure the display for use in a window created using the Tk framework. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_ENABLE | Specifies to configure the display for use in a window created using the Tk framework. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

The following inquire types are only available with windowed displays allocated on a remote system but displayed on the local computer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For displays allocated on a Distributed Aurora Imaging Library remote system but displayed on the local computer | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| InquireType | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| UserVarPtr<br>- Possible values returned | | | | | | | | | | | | | | | | |
| M_ASYNC_UPDATE | Inquires whether the data from the remote computer is transmitted to the local computer asynchronously or synchronously for display. SET | | | | | | | | | | | | | | | |
| *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_DEFAULT | Specifies to use the default mode, which can be selected using the Aurora Imaging Configurator utility. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DISABLE | Specifies to send display updates in synchronous mode. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_ENABLE | Specifies that data will be sent asynchronously. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COMPRESSION_TYPE | | [For essential Aurora Imaging Library Lite information, see remarks ]<br><br>Inquires the way in which data is compressed when transmitted from the remote computer to the local computer for display. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Specifies to use the default compression type, which can be selected using the Aurora Imaging Configurator utility. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_NULL | Specifies that compression is disabled. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_JPEG2000_LOSSLESS | Specifies that JPEG2000 lossless compression will be used. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_JPEG2000_LOSSY | Specifies JPEG2000 lossy compression will be used. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_JPEG_LOSSLESS | Specifies JPEG lossless compression will be used. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_JPEG_LOSSLESS_INTERLACED | Specifies that JPEG lossless compression will be used in separate fields. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_JPEG_LOSSY | Specifies that JPEG lossy compression will be used. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M_JPEG_LOSSY_INTERLACED | Specifies that JPEG lossy compression will be used in separate fields. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_Q_FACTOR | | [For essential Aurora Imaging Library Lite information, see remarks ]<br><br>Inquires the quantization factor used when compressing data transmitted from the remote computer to the local computer using a lossy compression. SET | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | *UserVarPtr extra info* | Data type info: Data type: *address of an AIL_INT* | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | M_DEFAULT | Specifies the default value; the default value is 80. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| | 1 <= *Value* <= 99 | Specifies the quantization factor. INFO | c | | e | f | g | a | | i | | j | k | l | m | n | b |

## Return value

The returned value is the requested information, cast to an *AIL_INT*. If the requested information does not fit into an *AIL_INT*, this function will return **M_NULL** or truncate the information.

## Remark

- *[Aurora Imaging Library Lite]*
  Note that during development and at runtime, compression support requires the presence of an Aurora Imaging Library license that grants access to the compression/decompression package. This access is only granted by default with the development license dongle for the full version of Aurora Imaging Library. In other cases, you must purchase access to this package separately.

## Compilation information

| | |
|---|---|
| **Header** | Include AIL.h. |
| **Library** | Use ail11.lib. |
| **DLL** | Requires ail11.dll; aildisplay11.dll. |

# MdispLut

## Synopsis

Associate a LUT buffer with a display.

## Syntax

```
void MdispLut(
    AIL_ID DisplayId, //out
    AIL_ID LutBufId   //in
)
```

## Description

This function associates a LUT buffer with the specified display. It can also be used to disassociate a LUT from the display.

If and when the display is selected (MdispSelect()), the change required to produce the display (LUT) effect occurs. Aurora Imaging Library checks the target display to determine whether or not a LUT is supported. If not, an error is generated.

When displaying using a LUT, note the following:

- If the LUT buffer values are changed while the image is selected on the display, the changes will not take effect until the next call is made to **MdispLut**(). That is, the LUT is not automatically updated when the LUT buffer is modified.
- You cannot select a 3-band image buffer to a display that is associated with a LUT buffer. In addition, the image buffer must be an 8- or 16-bit buffer. If these conditions are not respected, an error is generated.
- The view mode of the display cannot be set to M_AUTO_SCALE or M_MULTI_BYTES.
- The number of LUT buffer entries must be the same as the maximum number of intensities that can be represented in the displayed buffer. In other words, if you want to map an 8-bit 1-band image (that is, an image that can have 256 intensities), your LUT must also have 256 entries.

To disassociate a LUT buffer from a display, call **MdispLut**() with M_DEFAULT.

For more information, see the Lookup tables in general section of Chapter 24: Lookup tables and the Mapping 1-band images through a LUT upon display section of Chapter 25: Displaying an image.

## Parameters

DisplayId

Specifies the identifier of the display with which to associate the LUT buffer.

LutBufId INQ

Specifies a LUT buffer or predefined distinct colormap. This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For the identifier of the previously allocated LUT buffer | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| M_DEFAULT | Specifies that any association to a LUT buffer is removed from the specified display. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLORMAP_DISTINCT_256 | Specifies the LUT buffer with a distinct colormap as the LUT buffer to associate with the display. This colormap is designed to maximize the visual difference between consecutive colors. This colormap can be useful for colorizing images containing discrete objects, such as labeled blobs.<br><br>Note that this colormap is periodic, meaning that if the given LUT buffer has more than 256 entries, the remaining entries will be filled starting from the first color in the colormap.<br><br> | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLORMAP_GRAYSCALE | Specifies the LUT buffer with a grayscale colormap as the LUT buffer to associate with the display. This colormap transitions from black to white as the indices increase.<br><br> | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLORMAP_HOT | Specifies the LUT buffer with a hot colormap as the LUT buffer to associate with the display. This colormap transitions from black, to red, to yellow, and then, to white along the RGB cube as the indices increase. This colormap can be useful for displaying infrared images.<br><br> | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLORMAP_HUE | Specifies the LUT buffer with a hue colormap as the LUT buffer to associate with the display. This colormap is cyclic and transitions from red, to yellow, to green, to cyan, to blue, to magenta, and then, to red along the edge of the hue circle as the indices increase. This colormap is suitable to be repeated. This colormap is useful for displaying the hue component of an HSL image.<br><br> | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M_COLORMAP_JET | Specifies the LUT buffer with a jet colormap as the LUT buffer to associate with the display. This colormap transitions from dark blue, to blue, to cyan, to yellow, to red, and then, to dark red along the RGB cube as the indices increase. This colormap can be useful for displaying 3D elevation maps and thermal imaging. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLORMAP_SPECTRUM | Specifies the LUT buffer with a spectrum colormap as the LUT buffer to associate with the display. This colormap transitions from red, to yellow, to green, to blue, and then, to violet along the RGB cube as the indices increase. This colormap is a representation of the visual spectrum according to wavelengths. It can also be useful for displaying 3D elevation maps and thermal imaging. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLORMAP_TURBO | Specifies the LUT buffer with a turbo colormap as the LUT buffer to associate with the display. This colormap is an improved version of the jet colormap, with smoother color transitions. It transitions from dark blue, to blue, to cyan, to yellow, to red, and then, to dark red along the RGB cube as the indices increase. This colormap can be useful for displaying 3D elevation maps and thermal imaging. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_COLORMAP_WHEEL | Specifies the LUT buffer with a colorwheel colormap as the LUT buffer to associate with the display. This colormap is cyclic and transitions from blue, to magenta, to yellow, to green, and then, to blue again as the indices increase. This colormap is designed to be perceptually uniform (a given change in data is designed to be visually perceived as an equal change in color). This colormap is suitable to be repeated. This colormap can be useful for displaying 3D elevation maps. | c | | e | f | g | a | | i | | j | k | l | m | n | b |
| M_PSEUDO | Specifies the default pseudo-color LUT buffer as the LUT buffer to associate with the display. When this LUT is used and displaying an 8-bit image buffer, each gray intensity is displayed in a different color. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *LUT buffer identifier* | Specifies the identifier of the custom LUT buffer (allocated with MbufAlloc1d() or MbufAllocColor() with M_LUT) to associate with the display.<br><br>If you associate a three-band LUT buffer (RGB) with the display, and then select an image buffer to the display, the same image data is mapped through each band of the LUT, typically resulting in a color display effect (depends on the LUT values). | c | | e | f | g | a | | i | | j | k | l | m | n | b |

## Compilation information

| | |
|---|---|
| **Header** | Include AIL.h. |
| **Library** | Use ail11.lib. |
| **DLL** | Requires ail11.dll; aildisplay11.dll. |

# MdispPan

## Synopsis

Pan (scroll) a display.

## Syntax

```
void MdispPan(
    AIL_ID DisplayId,   //out
    AIL_DOUBLE XOffset, //in
    AIL_DOUBLE YOffset  //in
)
```

## Description

This function associates X- and Y-panning offsets with the specified display. When an image buffer is selected for display, the image is displaced on the display, from the top-left corner of the window (for windowed displays) or the screen (for exclusive displays), according to these offsets.

Note, the offsets (XOffset and YOffset) are in image pixels. Since the current zoom factors affect the displayed size of an image pixel, the panning offsets are also affected by the zoom factors. For example, if the display has an associated X-zoom factor of 4, panning by an X-offset of one image pixel results in panning by 4 pixels in the horizontal direction on the display.

To center the selected image buffer in the display, use MdispControl() with M_CENTER_DISPLAY, instead of using **MdispPan**().

For windowed and exclusive displays, you can also pan the display interactively using the Page-up/down and arrow keys, if the keys associated with the display have not been disabled using MdispControl() with M_KEYBOARD_USE. You can also pan the display by clicking on the image and dragging in the required direction with the mouse, if the mouse associated with the display has not been disabled using MdispControl() with M_MOUSE_USE. Note that these features are not enabled by default when using a user-defined window.

Note that the window must have the focus for the keys to work. To give focus to an exclusive display, you must permit the mouse cursor to move over the display, using MdispControl() with M_RESTRICT_CURSOR.

## Parameters

DisplayId

Specifies the identifier of the display.

XOffset INQ

Specifies the number of image pixels by which to pan an image buffer horizontally when it is displayed. Specify the offset relative to the top-left corner of the image buffer. Specify a positive offset value to displace the image to the left.

YOffset INQ

Specifies the number of image pixels by which to pan an image buffer vertically when it is displayed. Specify the offset relative to the top-left corner of the image buffer. Specify a positive offset value to

displace the image upwards.

## Compilation information

| | |
|---|---|
| **Header** | Include AIL.h. |
| **Library** | Use ail11.lib. |
| **DLL** | Requires ail11.dll; aildisplay11.dll. |

# MdispSelect

## Synopsis

Select an image buffer to display.

## Syntax

```
void MdispSelect(
    AIL_ID DisplayId, //out
    AIL_ID ImageBufId //in
)
```

## Description

This function outputs the content of the specified image buffer to the specified Aurora Imaging Library display. You can only display one image buffer at a time on a specific display.

For a windowed display, the image is presented in a window that has the same size as the image, unless such a window would not fit on the desktop or would be too small. If the image is too large to fit in the largest possible window given the selected display resolution, the top-left corner of the image will be aligned with the top-left corner of the largest possible window, and the right and bottom portion of the image, the part that exceeds the window area, will not be displayed; to view the missing portions of the image, you can pan the displayed image (see the Panning and zooming section of Chapter 25: Displaying an image for more information). If the image is smaller in size than the smallest possible window, the image will be centered in the smallest possible window, and the surrounding area will be filled with the background color (MdispControl() with M_BACKGROUND_COLOR). If the window is enlarged or maximized, the image will remain the same size and centered.

For an exclusive display, the image is presented full-screen, without a windowed border or frame, on one of the Windows desktop screens. The image is presented at the center of the screen. If the image is too large to fit on screen given the selected display resolution, the top-left corner of the image will be aligned with the top-left corner of the screen, and the right and bottom portion of the image, the part that exceeds the screen area, will not be displayed; to view the missing portions of the image, you can pan the displayed image (see the Panning and zooming section of Chapter 25: Displaying an image for more information). If the image is smaller in size than the screen area, the image will be centered on screen, and the surrounding area will be filled with the background color (MdispControl() with M_BACKGROUND_COLOR).

For an Aurora Imaging Web Library display, the image is presented in one or more connected instances of an Aurora Imaging Web Library client application (typically running on a remote computer). The size of the transmitted display is the same as the size of the image buffer. If the displayed image is panned or zoomed (see the Panning and zooming section of Chapter 25: Displaying an image for more information) so that it does not fill the display, the surrounding area will be filled with the background color (MdispControl() with M_BACKGROUND_COLOR). The Aurora Imaging Web Library client application might further crop or scale the displayed image after it is transmitted.

For an Aurora Imaging Library WPF display, the image will be displayed in the AILWPFControl with the bound display Id. If the control is larger than the selected image, the remainder of the control if filled with the display's background color (MdispControl() with M_BACKGROUND_COLOR). If the control is smaller than the selected image, the image will be aligned with the top-left corner of the control; to view the rest of the image, you can pan the displayed view (see the Panning and zooming section of Chapter 25: Displaying an image for more information).

You can control the displayed area using MdispControl(), MdispPan(), or MdispZoom(). For example, you can center the selected image in the display using MdispControl() with M_CENTER_DISPLAY.

You can remove an image buffer selected to the display using **MdispSelect**() with **M_NULL**. For windowed displays, this closes the associated window. For exclusive displays, this causes the display to disappear, allowing the desktop to reappear on the screen. For WPF, if the display was bound to a control, the content of the control is cleared with the display's background color (MdispControl() with M_BACKGROUND_COLOR).

You can only remove the entire image buffer from the display. Therefore, when displaying a parent buffer, you cannot remove one of its child buffers from the display.

Note that it is not necessary to stop displaying the image buffer before selecting another buffer for display.

You can annotate the image selected to the display non-destructively using the display's overlay mechanism (MdispControl() with M_OVERLAY) or using a 2D graphics list (MdispControl() with M_ASSOCIATED_GRAPHIC_LIST_ID). The display's overlay mechanism allows you to annotate the selected image using an automatically allocated, temporary, image buffer referred to as the display's overlay buffer; you can modify the overlay buffer with any function that writes to an image buffer (except a grabbing function). Annotating with a 2D graphics list allows you to annotate the selected image with the vector-based graphics in the 2D graphics list; you can add vector-based graphics to the 2D graphics list using Mgra...() and M...Draw() functions. Annotating using the latter technique avoids pixelization when the display is zoomed since the graphics are in vector format. For more information on non-destructive annotation see the Annotating the displayed image non-destructively section of Chapter 25: Displaying an image.

If the selected image is calibrated, the image is not physically corrected on display. If the display is associated with an overlay buffer (MdispControl() with M_OVERLAY), the camera calibration information of the selected buffer is copied to the overlay buffer; this allows you to draw in the overlay buffer in world units. If the camera calibration information of the overlay buffer is modified using McalAssociate(), MbufClear(), or some other function, the camera calibration information of the selected buffer will not be updated accordingly. However, changing the camera calibration information of the selected buffer will re-copy the camera calibration information of the selected buffer to the overlay buffer, even if the camera calibration information of the overlay buffer was manually changed.

## Parameters

DisplayId

Specifies the identifier of the display.

ImageBufId

Specifies the identifier of the image buffer to display. The image buffer must be a 1- or 3-band buffer. To be displayable, this buffer must be an image buffer that has an M_IMAGE + M_DISP attribute.

Set this parameter to **M_NULL** to stop displaying the currently selected image buffer on the specified display.

> Note that the image buffer and the display should be allocated on the same system.

## Compilation information

| | |
|---|---|
| **Header** | Include AIL.h. |

| | |
|---|---|
| **Library** | Use ail11.lib. |
| **DLL** | Requires ail11.dll; aildisplay11.dll. |

# MdispSelectWindow

## Synopsis

Select an image buffer to display in a user-defined window.

## Syntax

```
void MdispSelectWindow(
    AIL_ID DisplayId,                    //out
    AIL_ID ImageBufId,                   //in
    AIL_WINDOW_HANDLE ClientWindowHandle //in
)
```

## Description

This function outputs the content of the specified image buffer to the specified user-defined window, using the specified Aurora Imaging Library display.

If the specified image buffer is smaller in size than the target window size, the border outside the image is not modified. If the specified buffer is larger in size than the target window, the right and bottom portion of the buffer, the part that exceeds the window, is not displayed.

You can control the displayed area using MdispControl(), MdispPan(), or MdispZoom(). For example, you can center the selected image in the display using MdispControl() with M_CENTER_DISPLAY.

To remove an image buffer selected to the display using **MdispSelectWindow**(), you can use **MdispSelectWindow**() with **M_NULL**. This leaves the associated window open but leaves it blank.

Note that, to use a display with a window created using the third-party GTK, Tkinter, or Qt frameworks, you should enable a special mode. You can enable the mode using MdispControl() with M_GTK_MODE, M_TK_MODE, or M_QT_MODE, respectively. These modes prevent the display from appearing incorrectly when the window is moved or resized, while introducing a small amount of latency.

> This function only supports local windowed displays (MdispAlloc() with M_WINDOWED and without M_REMOTE_DISPLAY). To select an image buffer to all other display types, use MdispSelect().

## Parameters

DisplayId

Specifies the identifier of the display. The display must have been previously allocated using MdispAlloc with M_WINDOWED (and without M_REMOTE_DISPLAY).

ImageBufId

Specifies the image buffer to display. The image buffer must be a 1- or 3-band buffer. To be displayable, this buffer must be an image buffer that has an M_IMAGE + M_DISP attribute.

Set this parameter to **M_NULL** to stop displaying the currently selected image buffer on the specified display.

ClientWindowHandle

Specifies the window handle of the user-defined window or child window. This window must have a

window handle of type *HWND* or X11. For example, the Windows API functions can create a window with an *HWND* handle, and third-party functions (like Qt) can create an X11 window handle. If this parameter is set to zero, this function behaves like MdispSelect().

Set this parameter to **M_NULL** if the ImageBufId parameter is set to **M_NULL**.

## Compilation information

| | |
|---|---|
| **Header** | Include windows.h; AIL.h (windows.h must be included first). |
| **Library** | Use ail11.lib. |
| **DLL** | Requires ail11.dll; aildisplay11.dll. |

# MdispZoom

## Synopsis

Zoom a display.

## Syntax

```
void MdispZoom(
    AIL_ID DisplayId,   //out
    AIL_DOUBLE XFactor, //in
    AIL_DOUBLE YFactor  //in
)
```

## Description

This function associates a zoom factor in X and/or in Y with the specified display. When an image buffer is selected for display, it will be zoomed according to these factors. The image buffer is displayed starting from its top-left corner, unless panning is also associated with the display (using MdispPan()).

Instead of explicitly specifying a zoom factor, you can automatically scale the selected image buffer to fit the display, using MdispControl() with M_SCALE_DISPLAY. M_SCALE_DISPLAY ensures that the image aspect ratio is maintained. If you use M_SCALE_DISPLAY, the explicitly specified zoom factors (**MdispZoom**()) and pan settings (MdispPan()) will be overwritten.

For windowed and exclusive displays, you can also zoom the display interactively using the + and - keys, if the keys associated with the display have not been disabled using MdispControl() with M_KEYBOARD_USE.

Note that the window must have the focus for the keys to work. To give focus to an exclusive display, you must permit the mouse cursor to move over the display, using MdispControl() with M_RESTRICT_CURSOR.

Once you have permitted the mouse cursor to move over the display, you can use the mouse wheel to zoom in or out of the image interactively; to do so, scroll the wheel up or down, respectively.

## Parameters

DisplayId

Specifies the identifier of the display.

XFactor INQ

Specifies the zoom factor for the X-direction of the display. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For specifying the zoom factor in the X-direction | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| *Value* > 0.0 | Specifies the zoom factor for the X-direction of the display. A value greater than 1.0 will zoom in, while a factor less than 1.0 will zoom out. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

YFactor INQ

Specifies the zoom factor for the Y-direction of the display. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

| For specifying the zoom factor in the Y-direction | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Value | Description | (c) | (d) | (e) | (f) | (g) | (a) | (h) | (i) | (o) | (j) | (k) | (l) | (m) | (n) | (b) |
| *Value* > 0.0 | Specifies the zoom factor for the Y-direction of the display. A value greater than 1.0 will zoom in, while a factor less than 1.0 will zoom out. | c | | e | f | g | a | | i | | j | k | l | m | n | b |

## Compilation information

| | |
|---|---|
| **Header** | Include AIL.h. |
| **Library** | Use ail11.lib. |
| **DLL** | Requires ail11.dll; aildisplay11.dll. |